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SUBSTRATE PROCESSING SYSTEM AND TRANSFER METHOD

Abstract

A substrate processing system includes: a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring; a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and a controller. The controller performs: (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The application is a Bypass Continuation application of PCT International Application No. PCT/JP2023/038477, filed on Oct. 25, 2023 and designating the United States, the international application being based upon and claiming the benefit of priority from Japanese Patent Application No. 2022-174836, filed on Oct. 31, 2022, the entire content of each are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a substrate processing system and a transfer method.

BACKGROUND

[0003] Patent Document 1 discloses a plasma processing apparatus which performs a plasma processing on a substrate placed on a stage (substrate support) provided inside a processing chamber by disposing a focus ring (ring) at a periphery of the substrate support. A substrate processing system including the plasma processing apparatus performs an operation of, when exchanging the ring, taking out the ring from the substrate support by a transfer robot, cleaning a surface on which the ring is placed, and placing a ring on the substrate support again.

PRIOR ART DOCUMENTS

Patent Documents

[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. 2018-010992

SUMMARY

[0005] According to one embodiment of the present disclosure, there is provided a substrate processing system including: a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring; a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and a controller. The controller performs: (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure.

[0007] FIG. 1 is a view illustrating an example of a substrate processing system according to an

embodiment.

[0008] FIG. 2 is a schematic cross-sectional view illustrating an example of a plasma processing apparatus.

[0009] FIG. 3 is an enlarged view of a portion in FIG. 2.

[0010] FIG. 4 is a view illustrating a cause of a position misalignment which occurs in a ring when taking out the ring from a substrate support.

[0011] FIG. 5A is a view illustrating a first example of detecting an index related to a position misalignment amount of an inner ring. FIG. 5B is a view illustrating a second example of detecting an index related to a position misalignment amount of the inner ring. FIG. 5C is a view illustrating a third example of detecting an index related to a position misalignment amount of the inner ring.

[0012] FIG. 6 is a flowchart illustrating an operation sequence when taking out an edge ring.

[0013] FIG. 7 is a flowchart illustrating a processing flow of a misalignment identification and correction processing in a transfer method according to a first example.

[0014] FIG. 8A is a first view illustrating an operation of the misalignment identification and correction processing. FIG. 8B is a second view illustrating an operation continued from FIG. 8A. FIG. 8C is a third view illustrating an operation continued from FIG. 8B. FIG. 8D is a fourth view illustrating an operation continued from FIG. 8C. FIG. 8E is a fifth view illustrating an operation continued from FIG. 8D. FIG. 8F is a sixth view illustrating an operation continued from FIG. 8E. [0015] FIG. 9 is a flowchart illustrating a modification of the processing flow of the misalignment identification and correction processing according to the first example.

[0016] FIG. 10 is a flowchart illustrating a processing flow in a transfer method according to a second example.

[0017] FIG. 11 is a schematic cross-sectional view illustrating another example of the plasma processing apparatus.

[0018] FIG. 12 is a flowchart illustrating a processing flow in a transfer method according to a third example.

[0019] FIG. 13A is a first view illustrating a misalignment resolution operation. FIG. 13B is a second view illustrating an operation continued from FIG. 13A. FIG. 13C is a third view illustrating an operation continued from FIG. 13B.

[0020] FIG. 14 is a flowchart illustrating a processing flow in a transfer method according to a fourth example.

[0021] FIG. 15 is a flowchart illustrating a processing flow in a transfer method according to a fifth example.

[0022] FIG. 16 is a flowchart illustrating a processing flow in a transfer method according to a sixth example.

[0023] FIG. 17 is a view illustrating an example of imaging an inclined edge ring by a camera.

[0024] FIG. 18 is a view illustrating a modification of a configuration which performs a leakage check of gas as an index related to a position misalignment amount of the edge ring.

[0025] FIG. 19 is a flowchart illustrating a transfer operation sequence of the edge ring according to a modification.

[0026] FIG. 20A is a view illustrating a relationship between a position of the edge ring and positions of position detection sensors. FIG. 20B is a view illustrating a change in sensor output of the position detection sensors when transferring the edge ring from one position to the other position.

DETAILED DESCRIPTION

[0027] Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure.

However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures,

systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

[0028] Hereinafter, embodiments for implementing the present disclosure will be described with reference to the drawings. In the respective drawings, the same components may be denoted by the same reference numerals, and duplicated descriptions thereof may be omitted.

[Substrate Processing System]

[0029] A substrate processing system PS according to an embodiment is described with reference to FIG. 1. FIG. 1 is a view illustrating an example of the substrate processing system PS according to the embodiment. As illustrated in FIG. 1, the substrate processing system PS is a system capable of performing various types of processings, such as a plasma processing, on a substrate W. The substrate W may be, for example, a semiconductor wafer.

[0030] The substrate processing system PS includes a vacuum transfer module TM, a plurality of processing modules PM1 to PM7, a ring accommodation module RSM, a plurality of load lock modules LL1 to LL3, an atmospheric transfer module LM, load ports LP1 to LP4, an aligner AN, and a controller CU. The vacuum transfer module TM is also referred to as a transfer module. The processing modules PM1 to PM7 are also referred to as process modules. The ring accommodation module RSM is also referred to as a ring stocker module. The atmospheric transfer module LM is also referred to as a loader module.

[0031] The vacuum transfer module TM has a quadrangular shape in a plan view. The processing modules PM1 to PM7, the load lock modules LL1 to LL3, and the ring accommodation module RSM are connected to the vacuum transfer module TM. The vacuum transfer module TM includes a vacuum transfer chamber. The vacuum transfer chamber is maintained in a vacuum atmosphere. A transfer robot TR1 is installed in the vacuum transfer chamber (inside the vacuum transfer module TM).

[0032] The transfer robot TR1 is configured to be rotatable, extensible, and movable vertically. The transfer robot TR1 includes an upper fork FK1 and a lower fork FK2. The upper fork FK1 and the lower fork FK2 of the transfer robot TR1 are configured to be capable of holding each of the substrate W and a ring 113 (an inner ring 113a and an outer ring 113b). The transfer robot TR1 holds and transfers the substrate W and the ring 113 among the processing modules PM1 to PM7, the load lock modules LL1 to LL3, and the ring accommodation module RSM.

[0033] The upper fork FK1 is provided with a position detection sensor S1. The lower fork FK2 is provided with a position detection sensor S2. The position detection sensors S1 and S2 detect positions of the inner ring 113a and the outer ring 113b, which are loaded in the processing modules PM1 to PM7. The position detection sensors S1 and S2 may be, for example, optical displacement sensors, cameras or the like.

[0034] The vacuum transfer module TM may be provided with position detection sensors S11 and S12. The position detection sensors S11 and S12 are installed on a transfer path of the substrate W and the ring 113 (the inner ring 113a), which are transferred from the vacuum transfer module TM to the processing module PM1. The position detection sensors S11 and S12 are used when loading the substrate W or the ring 113 from the vacuum transfer module TM to the processing module PM1 and when unloading the substrate W or the ring 113 from the processing module PM1 to the vacuum transfer module TM. The position detection sensors S11 and S12 are installed, for example, in a vicinity of a gate valve (not illustrated) which partitions the vacuum transfer module TM and the processing module PM1. The position detection sensors S11 and S12 are disposed, for example, such that a distance between the position detection sensors S11 and S12 is smaller than an outer diameter of the substrate W and is smaller than an inner diameter of the inner ring 113a. Like the position detection sensors S11 and S12, the vacuum transfer module TM may be provided with position detection sensors S21, S22, S31, S32, S41, S42, S51, S52, S61, S62, S71, and S72.

[0035] The processing modules PM1 to PM7 are connected to the vacuum transfer module TM. The processing modules PM1 to PM7 have vacuum processing chambers. A substrate support 11

(see FIG. 2) is installed inside the vacuum processing chamber. After the substrate W is placed on the substrate support **11**, the processing modules PM**1** to PM**7** reduce the internal pressure, introduce a processing gas, and apply RF power applied to generate plasma, whereby a plasma processing is performed on the substrate W by the plasma. The vacuum transfer module TM is partitioned from the processing modules PM**1** to PM**7** by openable/closable gate valves (not illustrated).

[0036] The ring accommodation module RSM is an example of an apparatus which accommodates the ring **113**, and is connected to the vacuum transfer module TM. The ring accommodation module RSM accommodates, for example, the inner ring **113a** and the outer ring **113b**, which constitute the ring **113**. The ring accommodation module RSM may be configured to accommodate only the inner ring **113a**. The ring accommodation module RSM may be configured to accommodate only the outer ring **113b**. The inner ring **113a** and the outer ring **113b** are transferred between the processing modules PM**1** to PM**7** and the ring accommodation module RSM by the transfer robot TR**1**. The vacuum transfer module TM and the ring accommodation module RSM are partitioned by an openable/closable gate valve (not illustrated).

[0037] The load lock modules LL**1** to LL**3** are installed between the vacuum transfer module TM and the atmospheric transfer module LM. The load lock modules LL**1** to LL**3** are connected to the vacuum transfer module TM and the atmospheric transfer module LM. Each of the load lock modules LL**1** to LL**3** includes therein an internal pressure variable chamber of which an inside is switchable between vacuum and atmospheric pressure. The internal pressure variable chamber is provided with a stage (not illustrated) which enables the substrate W to be placed thereon. When transferring the substrate W from the atmospheric transfer module LM to the vacuum transfer module TM, the load lock modules LL**1** to LL**3** receive the substrate W from the atmospheric transfer module LM while maintaining the inside of the internal pressure variable chamber at the atmospheric pressure, and then depressurize the inside of the internal pressure variable chamber to transfer the substrate W to the vacuum transfer module TM. When transferring the substrate W from the vacuum transfer module TM to the atmospheric transfer module LM, the load lock modules LL**1** to LL**3** receive the substrate W from the vacuum transfer module TM while maintaining the inside of the internal pressure variable chamber in the vacuum, and then pressurize the inside of the internal pressure variable chamber to the atmospheric pressure to transfer the substrate W to the atmospheric transfer module LM. The load lock modules LL**1** to LL**3** are partitioned from the vacuum transfer module TM by openable/closable gate valves (not illustrated). The load lock modules LL**1** to LL**3** are partitioned from the atmospheric transfer module LM by openable/closable gate valves (not illustrated).

[0038] The atmospheric transfer module LM is installed to face the vacuum transfer module TM. The atmospheric transfer module LM may be, for example, an equipment front end module (EFEM). The atmospheric transfer module LM has a quadrangular shape in a plan view. The atmospheric transfer module LM includes an atmospheric transfer chamber. An inside of the atmospheric transfer chamber is maintained in an atmospheric pressure atmosphere. A transfer robot TR**2** is installed inside the atmospheric transfer chamber. The transfer robot TR**2** holds and transfers the substrate W among the load ports LP**1** to LP**4**, the aligner AN, and the load lock modules LL**1** to LL**3**. The atmospheric transfer module LM may have a fan filter unit (FFU).

[0039] The load ports LP**1** to LP**4** are connected to the atmospheric transfer module LM. A plurality of substrate accommodation containers CS**1** is placed in the load lock ports LP**1** to LP**4**. The substrate accommodation container CS**1** may be, for example, a front-opening unified pod (FOUP) which accommodates a plurality of substrates W (for example, 25 substrates W).

[0040] The aligner AN is connected to the atmospheric transfer module LM. The aligner AN is configured to perform position adjustment of the substrate W. The aligner AN may be installed inside the atmospheric transfer chamber.

[0041] The controller CU controls each part of the substrate processing system PS. The controller

CU controls, for example, an operation of the transfer robot TR1 installed in the vacuum transfer module TM, an operation of the transfer robot TR2 installed in the atmospheric transfer module LM, and opening and closing of the gate valves. The controller CU may be, for example, a computer. The controller CU includes a central processing unit (CPU) which is a processor, a random access memory (RAM), a read only memory (ROM), an auxiliary storage device, and the like. The CPU operates based on a program stored in the ROM or the auxiliary storage device, and controls each part of the substrate processing system PS.

[Plasma Processing Apparatus]

[0042] An example of a plasma processing apparatus **1** applied to the processing modules PM1 to PM7 in FIG. **1** is described with reference to FIGS. **2** and **3**. FIG. **2** is a schematic cross-sectional view illustrating an example of the plasma processing apparatus **1**. FIG. **3** is an enlarged view of a portion in FIG. **2**.

[0043] The plasma processing apparatus **1** includes a plasma processing chamber **10** (processing chamber), a gas supply **20**, an RF power supply **30**, an exhaust system **40**, a lifter **50**, and a controller **90**.

[0044] The plasma processing chamber **10** includes a substrate support **11** and an upper electrode **12**. The substrate support **11** is disposed in a lower region of a plasma processing space **10s** in the plasma processing chamber **10**. The upper electrode **12** is disposed above the substrate support **11**, to function as a portion of a ceiling plate of the plasma processing chamber **10**.

[0045] The substrate support **11** supports the substrate W in the plasma processing space **10s**. The substrate support **11** includes a lower electrode **111**, an electrostatic chuck **112**, a ring **113** (hereinafter, also referred to as a ring assembly **113**), and an insulating member **115**.

[0046] The electrostatic chuck **112** is disposed on the lower electrode **111**. The electrostatic chuck **112** has an upper surface which includes a substrate support surface **112a** and a ring support surface **112b**. The electrostatic chuck **112** supports the substrate W on the substrate support surface **112a**. The electrostatic chuck **112** supports the inner ring **113a** on the ring support surface **112b**. The electrostatic chuck **112** includes an insulating member **112c**, a first attraction electrode **112d**, and a second attraction electrode **112e**. The first attraction electrode **112d** and the second attraction electrode **112e** are embedded in the insulating member **112c**. The first attraction electrode **112d** is located below the substrate support surface **112a**. The electrostatic chuck **112** attracts and holds the substrate W on the substrate support surface **112a** by applying a voltage to the first attraction electrode **112d**. The second attraction electrode **112e** is located below the ring support surface **112b**. The electrostatic chuck **112** attracts and holds the inner ring **113a** on the ring support surface **112b** by applying a voltage to the second attraction electrode **112e**. In the examples in FIGS. **2** and **3**, the electrostatic chuck **112** includes a monopolar electrostatic chuck that attracts and holds the substrate W, and a bipolar electrostatic chuck that attracts and holds the inner ring **113a**. A bipolar electrostatic chuck may be used instead of the monopolar electrostatic chuck, and a monopolar electrostatic chuck may be used instead of the bipolar electrostatic chuck.

[0047] The ring assembly **113** includes the inner ring **113a** and the outer ring **113b**. The inner ring **113a** has an annular shape. The inner ring **113a** is placed on the ring support surface **112b** to surround the substrate W. The inner ring **113a** improves uniformity of a plasma processing on the substrate W. The inner ring **113a** is made of, for example, a conductive material such as silicon (Si) or silicon carbide (SiC). Further, the inner ring **113a** may be made of an insulating material such as quartz. The outer ring **113b** has an annular shape. The outer ring **113b** is disposed on an outer peripheral portion of the inner ring **113a**. The outer ring **113b** protects an upper surface of the insulating member **115** from, for example, plasma. The outer ring **113b** is made of, for example, an insulating material such as quartz. Further, the outer ring **113b** may be made of a conductive material such as silicon or silicon carbide. In the illustrated example, an inner peripheral portion of the outer ring **113b** is located inward of the outer peripheral portion of the inner ring **113a**, the outer peripheral portion of the inner ring **113a** is located outward of the inner peripheral portion of the

outer ring **113b**, and the inner ring **113a** and the outer ring **113b** partially overlap each other in top view. Accordingly, when a plurality of supporting pins **521** to be described later is raised and lowered, the outer ring **113b** and the inner ring **113a** are raised and lowered. The insulating member **115** is disposed to surround the lower electrode **111**. The insulating member **115** is fixed to a bottom portion of the plasma processing chamber **10**, to support the lower electrode **111**.

[0048] The upper electrode **12** along with an insulating member **13** constitutes the plasma processing chamber **10**. The upper electrode **12** supplies one or more types of processing gases from the gas supply **20** to the plasma processing space **10s**. The upper electrode **12** includes a ceiling plate **121** and a support **122**. A lower surface of the ceiling plate **121** defines the plasma processing space **10s**. A plurality of gas introduction ports **121a** is formed in the ceiling plate **121**. Each of the plurality of gas introduction ports **121a** penetrates through the ceiling plate **121** in a plate thickness direction (vertical direction) of the ceiling plate **121**. The support **122** detachably supports the ceiling plate **121**. A gas diffusion chamber **122a** is provided inside the support **122**. A plurality of gas introduction ports **122b** extends downward from the gas diffusion chamber **122a**. The plurality of gas introduction ports **122b** communicates with the plurality of gas introduction ports **121a**, respectively. The support **122** is provided with a gas supply port **122c**. The upper electrode **12** supplies one or more processing gases from the gas supply port **122c** to the plasma processing space **10s** via the gas diffusion chamber **122a**, the plurality of gas introduction ports **122b**, and the plurality of gas introduction ports **121a**.

[0049] A sidewall of the plasma processing chamber **10** is provided with a loading/unloading port **10p**. The substrate W is transferred between the plasma processing space **10s** and an outside of the plasma processing chamber **10** via the loading/unloading port **10p**. The loading/unloading port **10p** is opened and closed by a gate valve G.

[0050] The gas supply **20** includes one or more gas sources **21** and one or more flow rate controllers **22**. The gas supply **20** supplies one or more types of processing gases from the respective gas sources **21** to the gas supply port **122c** via the respective flow rate controllers **22**. The flow rate controller **22** may include, for example, a mass flow controller or a pressure-controlled flow rate controller. The gas supply **20** may include one or more flow rate modulation devices that modulate or pulse flow rates of one or more processing gases.

[0051] The RF power supply **30** includes two RF power sources (a first RF power source **31a** and a second RF power source **31b**) and two matchers (a first matcher **32a** and a second matcher **32b**). The first RF power source **31a** supplies first RF power to the lower electrode **111** via the first matcher **32a**. A frequency of the first RF power may be, for example, 13 MHz to 150 MHz. The second RF power source **31b** supplies second RF power to the lower electrode **111** via the second matcher **32b**. A frequency of the second RF power may be, for example, 400 kHz to 13.56 MHz. Instead of the second RF power source **31b**, a DC power source may be used.

[0052] The exhaust system **40** may be connected to, for example, a gas exhaust port **10e** provided at the bottom portion of the plasma processing chamber **10**. The exhaust system **40** may include a pressure adjusting valve and a vacuum pump. A pressure in the plasma processing space **10s** is adjusted by the pressure adjusting valve. The vacuum pump may include a turbo molecular pump, a dry pump, or a combination thereof.

[0053] The lifter **50** includes a first lifter **51** and a second lifter **52**.

[0054] The first lifter **51** includes a plurality of supporting pins **511** and an actuator **512**. The plurality of supporting pins **511** is inserted into through-holes H1 formed in the lower electrode **111** and the electrostatic chuck **112** so as to be capable of protruding and retracting from an upper surface of the electrostatic chuck **112**. The plurality of supporting pins **511** protrudes from the upper surface of the electrostatic chuck **112** to support the substrate W with upper ends of the plurality of supporting pins **511**, which are in contact with a lower surface of the substrate W. The actuator **512** raises and lowers the plurality of supporting pins **511**. As the actuator **512**, for example, a motor such as a DC motor, a stepping motor, or a linear motor, an air driving

mechanism such as an air cylinder, or a piezoelectric actuator may be used. For example, when transferring the substrate W between the transfer robot TR1 and the substrate support 11, the first lifter 51 raises and lowers the plurality of supporting pins 511.

[0055] The second lifter 52 includes the plurality of supporting pins 521 and an actuator 522. The supporting pin 521 is a stepped supporting pin formed of cylindrical (solid rod-shaped) members. The supporting pin 521 has a lower pin 523 and an upper pin 524. The upper pin 524 is installed on the lower pin 523. An outer diameter of the lower pin 523 is larger than an outer diameter of the upper pin 524. Accordingly, a stepped portion is formed by an upper end surface 523a of the lower pin 523. The lower pin 523 and the upper pin 524 are, for example, integrally formed.

[0056] The supporting pin 521 is inserted into a through-hole H11 formed in the lower electrode 111, a through-hole H12 formed in the insulating member 115, and a through-hole H13 formed in the outer ring 113b so as to be capable of protruding and retracting from the upper surface of the insulating member 115 and an upper surface of the outer ring 113b. Inner diameters of the through-holes H11 and H12 are slightly larger than the outer diameter of the lower pin 523. An inner diameter of the through-hole H13 is slightly larger than the outer diameter of the upper pin 524, and is smaller than the outer diameter of the lower pin 523.

[0057] The supporting pin 521 may be displaced between a standby position, a first support position, and a second support position.

[0058] The standby position is a position at which an upper end surface 524a of the upper pin 524 is below a lower surface of the inner ring 113a. When the supporting pin 521 is in the standby position, the inner ring 113a and the outer ring 113b are supported on the electrostatic chuck 112 and the insulating member 115, respectively, without being lifted by the supporting pin 521.

[0059] The first support position is a position above the standby position. The first support position is a position at which the upper end surface 524a of the upper pin 524 protrudes above the upper surface of the outer ring 113b, and the upper end surface 523a of the lower pin 523 is below a lower surface of the outer ring 113b. As the supporting pin 521 moves to the first support position, the upper end surface 524a of the upper pin 524 comes into contact with a recess formed in the lower surface of the inner ring 113a to support the inner ring 113a.

[0060] The second support position is a position above the first support position. The second support position is a position at which the upper end surface 523a of the lower pin 523 protrudes above the upper surface of the insulating member 115. As the supporting pin 521 moves to the second support position, the upper end surface 524a of the upper pin 524 comes into contact with the recess to support the inner ring 113a, and the upper end surface 523a of the lower pin 523 comes into contact with the lower surface of the outer ring 113b to support the outer ring 113b.

[0061] The actuator 522 raises and lowers the plurality of supporting pins 521. The actuator 522 may be configured similarly to the actuator 512.

[0062] When transferring the inner ring 113a between the transfer robot TR1 and the substrate support 11, the second lifter 52 lifts the inner ring 113a by moving the plurality of supporting pins 521 to the first support position. When transferring the inner ring 113a and the outer ring 113b between the transfer robot TR1 and the substrate support 11, the second lifter 52 lifts the inner ring 113a and the outer ring 113b by moving the plurality of supporting pins 521 to the second support position. Alternatively, even when transferring the outer ring 113b between the transfer robot TR1 and the substrate support 11 in a state in which the inner ring 113a does not exist, the second lifter 52 lifts the outer ring 113b by moving the plurality of supporting pins 521 to the second support position.

[0063] The controller 90 controls each part of the plasma processing apparatus 1. The controller 90 includes, for example, a computer 91. The computer 91 includes, for example, a CPU 911 which is a processor, a storage 912, and a communication interface 913. The CPU 911 is configured to perform various control operations, based on a program stored in the storage 912. The storage 912 includes at least one memory type selected from a group consisting of auxiliary storage devices

such as a RAM, a ROM, a hard disk drive (HDD), and a solid state drive (SSD). The communication interface **913** may communicate with the plasma processing apparatus **1** via a communication line such as a local area network (LAN). The controller **90** may be provided separately from the controller CU or may be included in the controller CU.

(Position Misalignment of Ring)

[0064] Next, a cause of a position misalignment which occurs in the ring **113** when taking out the ring **113** from the substrate support **11** is described with reference to FIG. **4**.

[0065] In the processing modules PM**1** to PM**7** (e.g., the plasma processing apparatus **1** in FIG. **2**), when processing the substrate W, the substrate W is electrostatically attracted and fixed by the electrostatic chuck **112**. Further, in the plasma processing apparatus **1**, at a periphery of the substrate W, the ring **113** (the inner ring **113a**) is electrostatically attracted and fixed by the electrostatic chuck **112**.

[0066] In the plasma processing apparatus **1**, after a plasma processing, charges in the electrostatic chuck **112** are reduced by a charge removal processing (gas charge removal, plasma charge removal, or the like). Further, by applying a direct current voltage having a polarity different from that in the plasma processing to the electrostatic chuck **112** of the substrate support **11**, electrostatic forces of the substrate support surface **112a** and the ring support surface **112b** may be reduced.

[0067] Although the charge removal processing is performed, charges may slightly remain in the substrate support surface **112a** and the ring support surface **112b**. Therefore, even after the charge removal processing, attraction of the inner ring **113a** to the ring support surface **112b** may slightly occur (hereinafter, also referred to as residual attraction). When the inner ring **113a** is spaced apart from the ring support surface **112b** by raising the supporting pins **521** so as to exchange the inner ring **113a**, the inner ring **113a** vibrates due to the residual attraction of the inner ring **113a**, and hence the inner ring **113a** is placed on the supporting pins **521** in a state in which the inner ring **113a** is misaligned from a normal position of the inner ring **113a**. Accordingly, there occurs a phenomenon in which the inner ring **113a** is placed in a position misaligned from an original holding position (the supporting pins **521**) at which the transfer robot TR**1** holds the inner ring **113a**.

[0068] For example, as illustrated in the upper right drawing of FIG. **4**, the inner ring **113a** may be misaligned in a horizontal direction (lateral direction). Accordingly, when transferring the inner ring **113a** to the transfer robot TR**1**, the transfer robot TR**1** holds the inner ring **113a** at a misaligned position. The misaligned inner ring **113a** may cause a problem such as that, in transfer of the transfer robot TR**1**, the inner ring **113a** is broken due to interference with other parts.

[0069] Further, for example, as illustrated in the lower right drawing of FIG. **4**, the position of the inner ring **113a** is considerably misaligned due to the residual attraction, and therefore, the inner ring **113a** may be obliquely inclined while getting out of some supporting pins **521** among the supporting pins **521**. In this case, the transfer robot TR**1** which slides in the horizontal direction above the substrate support **11** collides with the inner ring **113a**, and therefore, the transfer robot TR**1** and the inner ring **113a** may be broken. That is, in the substrate processing system PS, it is required for the transfer robot TR**1** to perform unloading or loading by suppressing the position misalignment of the inner ring **113a**. However, in a conventional substrate processing system, when taking output an inner ring (edge ring) from a substrate support, whether a transfer robot held the inner ring at a normal position was not recognized.

[0070] When unloading the inner ring **113a**, the controller CU of the substrate processing system PS according to this embodiment acquires an index related to a position misalignment amount of the inner ring **113a**. Further, when the position of the inner ring **113a** is misaligned, the controller CU performs correction for resolving the position misalignment. Next, a configuration of acquiring an index related to a position misalignment amount of the inner ring **113a** is described with reference to FIGS. **5A** to **5C**.

[0071] FIG. **5A** is a view illustrating a first example of detecting an index related to a position

misalignment amount of the inner ring **113a**. The substrate processing system PS uses, as the index related to the position misalignment amount, a detection result of a position of the inner ring **113a** detected by the position detection sensor **S1** installed on the upper fork **FK1** (or the position detection sensor **S2** installed on the lower fork **FK2**). The position detection sensors **S1** and **S2** may be installed on a lower surface of the upper fork **FK1** or the lower fork **FK2**, which faces the substrate support **11** (a surface opposite to a surface on which the substrate **W** is held). Further, a plurality of position detection sensors **S1** or **S2** may be installed at the upper fork **FK1** or the lower fork **FK2**. For example, the position detection sensor **S1** may be installed at each of two prong portions of the upper fork **FK1**. Further, an installation place of the position detection sensor **S1** or **S2** is not limited to the lower surface facing the substrate support **11** (the surface opposite to the surface on which the substrate **W** is held), and the position detection sensor **S1** or **S2** may be installed at a side surface of the upper fork **FK1** or the lower fork **FK2**.

[0072] For example, an optical detector (e.g., a displacement sensor) capable of detecting a shape change of an object may be applied to the position detection sensors **S1** and **S2**. The position detection sensor **S1** or **S2** detects an edge of the substrate support surface **112a** by detecting a step difference between the substrate support surface **112a** and the ring support surface **112b** along with movement of the upper fork **FK1** or the lower fork **FK2** in the horizontal direction (left arrow of **S1** in FIG. 5A). Further, an upper surface of the inner ring **113a** includes an inner portion **113a1** lower than the substrate support surface **112a** on an inner side in a radial direction and an outer portion **113a2** higher than the inner portion **113a1** on an outer at a diameter direction outer side of the inner portion **113a1**. The position detection sensor **S1** or **S2** detects an edge of the inner ring **113a** by detecting a step difference between the inner portion **113a1** and the outer portion **113a2** along with movement of the upper fork **FK1** or the lower fork **FK2** in the horizontal direction (right arrow of **S1** in FIG. 5A). The controller CU may obtain a gap amount between the substrate support surface **112a** and the inner ring **113a**, based on a horizontal position (X coordinate and Y coordinate) of the edge of the substrate support surface **112a** and a horizontal position (X coordinate and Y coordinate) of the edge of the inner ring **113a**. The gap amount is a length of a gap in the horizontal direction between an inner peripheral surface of the inner portion **113a1** of the inner ring **113a** and a sidewall between the substrate support surface **112a** and the ring support surface **112b**. For example, the gap amount is calculated as a value decomposed into an amount in an X-axis direction and an amount in a Y-axis direction.

[0073] The gap amount is specified in advance according to an outer diameter of the substrate support surface **112a** and an inner diameter of the inner ring **113a**. When there exists a portion of an annular gap, at which an absolute value of the gap amount is larger than a specified value, a position misalignment amount of the inner ring **113a** becomes large. As such, the controller CU may calculate the position misalignment amount of the inner ring **113a** with a high degree of precision, based on the detection result of the position detection sensor **S1** or **S2** (an example of the index related to the position misalignment amount).

[0074] FIG. 5B is a view illustrating a second example of detecting an index related to a position misalignment amount of the inner ring **113a**. The substrate processing system PS may perform a gas leakage check between the ring support surface **112b** and the inner ring **113a** as the index related to the position misalignment amount.

[0075] That is, a gas supply port **61a** is formed in the ring support surface **112b** at a peripheral edge of the electrostatic chuck **112**. The gas supply port **61a** supplies a gas into a gap between a rear surface of the inner ring **113a** placed on the ring support surface **112b** and the ring support surface **112b**. The gas may be the same as a heat transfer gas supplied to a bottom surface of the inner ring **113a** in a plasma processing. As an example of the gas, He gas may be applied. Further, in a gas flow path **61** that communicates with the gas supply port **61a**, an end portion of the gas flow path **61**, which is opposite to the ring support surface **112b**, is connected to a gas supply **66** via a pipe **62**. The gas supply **66** may include one or more gas sources **661** and one or more flow rate

controllers **662**. In an embodiment, the gas supply **66** is configured to, for example, supply the gas from the gas source **661** to the gas supply port **61a** via the flow rate controller **662**. Each flow rate controller **662** may include, for example, a mass flow controller or a pressure-controlled flow rate controller. The gas flow path **61** and the pipe **62** may function as at least a portion of a supply path that supplies the gas between the ring support surface **112b** and the rear surface of the inner ring **113a**.

[0076] Further, the end portion of the gas flow path **61**, which is opposite to the ring support surface **112b**, is connected to an exhaust system **64** via the pipe **62**. Accordingly, a periphery of the ring support surface **112b** of the electrostatic chuck **112** may be exhausted via the gas supply port **61a**. That is, the gas supply port **61a** may function as an exhaust hole that exhausts the periphery of the ring support surface **112b**. Therefore, in the embodiment, the gas flow path **61** and the pipe **62** may function as at least a portion of an exhaust path that exhausts the gap between the ring support surface **112b** and the rear surface of the inner ring **113a**.

[0077] Further, as for the electrostatic chuck **112**, a pressure sensor **67** that measures a pressure of the gap between the inner ring **113a** electrostatically attracted to the ring support surface **112b** and the ring support surface **112b** is provided. The pressure sensor **67** is, for example, installed in the pipe **62**. Further, a switching valve **65** that switches execution and stop of gas supply by the gas supply **66** may be installed in the pipe **62**. Similarly, a switching valve **63** that switches execution and stop of exhaust of the periphery of the ring support surface **112b** by the exhaust system **64** may be installed in the pipe **62**.

[0078] In a gas leakage check, the controller CU of the substrate processing system PS operates, for example, a gas check mechanism **60** (which including the gas flow path **61**, the pipe **62**, the switching valve **63**, the exhaust system **64**, the switching valve **65**, the gas supply **66** and the pressure sensor **67**) and the electrostatic chuck **112** according to a sequence described below. First, the controller CU applies a voltage to the second attraction electrode **112e** of the electrostatic chuck **112** in a state in which the substrate W does not exist in the plasma processing chamber **10**, the inner ring **113a** is placed on the ring support surface **112b**, and exhaust is performed via the gas flow path **61** by the exhaust system **64**. Accordingly, a direct current voltage (e.g., direct current voltages having different polarities in the case of bipolar second attraction electrodes **112e**) is applied to the second attraction electrode **112e** of the electrostatic chuck **112**.

[0079] Subsequently, the controller CU supplies a gas to the gas flow path **61** such that a pressure in the gas flow path **61** is maintained higher than a pressure in the plasma processing chamber **10**. Specifically, the controller CU closes the switching valve **63** and stops the exhaust via the gas flow path **61** by the exhaust system **64**. Meanwhile, the controller CU opens the switching valve **65** and supplies the gas by the gas supply **66**. The gas is supplied into a gap between the rear surface of the inner ring **113a** and the ring support surface **112b** via the pipe **62** and the gas flow path **61**. If a pressure in the gap becomes a target pressure (e.g., if a result measured by the pressure sensor **67** becomes the target pressure), the controller CU closes the switching valve **65** and stops the supply of the gas. The target pressure is, for example, the same as a pressure in the gap in the plasma processing.

[0080] After that, the controller CU measures a pressure in a flow path including the gas flow path **61** by the pressure sensor **67**. Specifically, the pressure sensor **67** measures a pressure in the pipe **62** after a predetermined time elapses from the stop of the supply of the gas. The measured pressure is substantially the same as the pressure in the gap between the ring support surface **112b** and the inner ring **113a**. Therefore, the controller CU may determine leakage of the gas from the gap (an index related to a position misalignment amount of the inner ring **113a**), based on the pressure measured by the pressure sensor **67**. More specifically, the controller CU determines whether the measured pressure is less than a pressure threshold value as determination of gas leakage from the gap. The pressure threshold value is, for example, set to 90% to 98% of the target pressure, and information of the pressure threshold value is stored in advance in the storage **912**. If the measured

pressure is less than the pressure threshold value, the controller CU determines that the gas is leaked from the gap between the ring support surface **112b** and the inner ring **113a**, i.e., that the position of the inner ring **113a** is misaligned.

[0081] FIG. 5C is a view illustrating a third example of detecting an index related to a position misalignment amount of the inner ring **113a**. The controller CU acquires, as the index related to the position misalignment amount, image information of a camera CM installed on the upper fork FK1 or a camera CM installed on the lower fork FK2. The camera CM is, for example, installed on the lower surface of the upper fork FK1 or the lower fork FK2, which faces the substrate support **11**, instead of the position detection sensors S1 and S2. Further, the camera CM may be installed on a side surface of the upper fork FK1 or the lower fork FK2.

[0082] In the image information, for example, a gap between the edge of the substrate support surface **112a** and the edge of the inner ring **113a** is included as color or contrast information. Therefore, the controller CU acquires image information imaged by the camera CM (an example of the index related to the position misalignment amount), and performs an appropriate image processing on the image information, so that it is possible to calculate a gap amount (position misalignment amount) with a high degree of precision.

[0083] Back to FIG. 1, the controller CU of the substrate processing system PS determines necessity of exchange of the inner ring **113a**, based on a trigger such as an instruction by a user, a number of substrate processings, quality of the substrate W, a sensor value of each of the processing modules PM1 to PM7, or error occurrence. When determining that the exchange is necessary, the controller CU collects the inner ring **113a** of the processing module for an exchange target, and performs a processing of exchanging the inner ring **113a** with an inner ring **113a** for exchange, which is accommodated in the ring accommodation module RSM. The inner ring **113a** for exchange may be a new product (not used) or may be one that is completely used but is not consumed so much.

[0084] Further, when taking out the inner ring **113a** from the processing module for the exchange target, the controller CU acquires an index related to a position misalignment amount of the inner ring **113a** as described above to recognize whether a misalignment has occurred in the inner ring. Accordingly, it is possible to for the substrate processing system PS to correct the misalignment of the inner ring **113a**, and thus it is possible to avoid damage of a part and the like.

[Transfer Method]

[0085] FIG. 6 is a flowchart illustrating an operation sequence when taking out an edge ring ER. Subsequently, an operation until taking out the edge ring ER is described with reference to FIG. 6. The edge ring ER corresponds to the inner ring **113a** illustrated in FIGS. 3, 4 and 5A to 5C. Hereinafter, a case where the inner ring **113a** is transferred between the processing module PM1 and the ring accommodation module RSM is described. Even in a case where a processing module for an exchange target is any one of the processing modules PM2 to PM7, the same method as a case where the processing module for the exchange target is the processing module PM1 may be used.

[0086] The operation sequence includes steps S1 to S6. The steps S1 to S6 are performed as the controller CU and/or the controller **90** controls each part of the substrate processing system PS. Although the controller CU is described separately from the controller **90**, the controller CU may have functions of the controller **90** to control all processings, and the controller **90** may have functions of the controller CU to control all processings.

[0087] In the step S1, the controller CU controls the plasma processing apparatus **1** constituting the processing module PM1 to perform a plasma processing (substrate processing) on the substrate W in the plasma processing chamber **10**. In the plasma processing, the controller **90** of the plasma processing apparatus **1** attracts the substrate W and the edge ring ER by applying a direct current voltage to the first attraction electrode **112d** and the second attraction electrode **112e** of the electrostatic chuck **112**.

[0088] Then, after the plasma processing, the controller CU unloads the substrate W in the plasma processing apparatus **1** by the transfer robot TR**1**. At this time, the controller **90** stops the application of the direct current voltage to the first attraction electrode **112d** (or applies another direct current voltage having a polarity different from a polarity of the direct current voltage) and enables the substrate W to be taken out. The controller **90** raises the substrate W by the first lifter **51** and transfers the substrate W to the transfer robot TR**1** which enters into the plasma processing chamber **10**. The transfer robot TR**1** unloads the substrate W from the plasma processing apparatus **1**, and transfers the substrate W to a next processing module or any one of the load lock modules LL**1** to LL**3**.

[0089] In the step S2, after unloading the substrate W, the controller CU controls the plasma processing apparatus **1** to operate in an idle mode. The idle mode is a mode after the substrate W is plasma-processed, and is a mode in which the plasma processing is not performed. In the idle mode, the controller CU determines necessity of exchange of the edge ring ER, and determines whether the exchange of the edge ring ER can be performed. Necessity of exchange of the ring **113** may be determined based on the above-described trigger. The controller CU determines whether the exchange can be performed, based on existence of the substrate W in the processing module PM**1** for the exchange target, a transfer schedule of the transfer robot TR**1**, or the like. Further, in the step S2, the controller **90** keeps attracting the inner ring **113a** by applying the direct current voltage to the second attraction electrode **112e**. The determination of the exchange of the edge ring ER is not limited to that in the idle mode, and may be performed at another timing such as a plasma processing.

[0090] When determining that the exchange of the inner ring **113a** is performed, in the step S3, the controller CU controls the plasma processing apparatus **1** to perform a charge removal processing on the inner ring **113a**. For example, when performing a gas charge removal processing in the charge removal processing, the plasma processing apparatus **1** controls the plasma processing chamber **10** at a predetermined pressure by supplying an inert gas such as N₂ gas into the plasma processing chamber **10** by the gas supply **20**, and exhausting a gas in the plasma processing chamber **10** by the exhaust system **40**. Further, the plasma processing apparatus **1** applies a direct current voltage having a polarity different from a polarity of the direct current voltage applied in the plasma processing to the second attraction electrode **112e** of the electrostatic chuck **112** for a predetermined time, and then stops. Further, the plasma processing apparatus **1** stops the pressure control and ends the gas charge removal.

[0091] Further, in the steps S4 to S6, the controller CU transfers the edge ring ER from the processing module PM**1** to the ring accommodation module RSM. First, a case where state identification and transfer of the edge ring ER is performed by the position detection sensor S**1** or S**2** of the transfer robot TR**1** illustrated in FIG. 5A (hereinafter, also referred to as a transfer method according to a first example) is described.

[0092] In the step S4, in a case where the controller CU identifies a misalignment of the inner ring **113a** when raising the inner ring **113a** from the substrate support **11**, the controller CU performs a misalignment identification and correction processing of correcting the misalignment.

[0093] FIG. 7 is a flowchart illustrating a processing flow of the misalignment identification and correction processing. FIGS. 8A to 8F are views illustrating an operation of the misalignment identification and correction processing. In the misalignment identification and correction processing, first, the controller CU raises the supporting pins **521** of the plasma processing apparatus **1** and separates the edge ring ER from the ring support surface **112b** of the substrate support **11** (step S101 in FIG. 7). When residual attraction exists between the ring support surface **112b** and the edge ring ER, a misalignment may occur in the edge ring ER when raised as illustrated in FIG. 8A.

[0094] The controller CU lowers the supporting pins **521** of the plasma processing apparatus **1** and temporarily places the edge ring ER on the ring support surface **112b** of the substrate support **11** for

the time being (step **S102** in FIG. 7). When the misalignment of the edge ring ER occurs, the edge ring ER is disposed while being misaligned from the substrate support surface **112a** as illustrated in FIG. 8B.

[0095] Subsequently, the controller CU operates the transfer robot TR1 and detects a position of the edge ring ER (an example of an index related to a position misalignment amount) by the position detection sensor S1 or S2 (step **S103** in FIG. 7). For example, as illustrated in FIG. 8C, the transfer robot TR1 moves in the horizontal direction within the plasma processing space **10s** to dispose the position detection sensor S1 or S2 at a position capable of detecting an edge of the substrate support surface **112a** and an edge of the edge ring ER. Accordingly, the controller CU receives a detection result of the position detection sensor S1 or S2, thereby obtaining gap amounts (position misalignment amounts) of the substrate support surface **112a** and the edge ring ER.

[0096] Subsequently, the controller CU determines whether the position misalignment amount of the edge ring ER is a pre-stored threshold value or more by comparing the position misalignment amount of the edge ring ER with the threshold value (step **S104**). The threshold value may be set to an appropriate value according to a measurement of the substrate support surface **112a** or the edge ring ER, and may be set to, for example, a value in a range of about 0.1 mm to about 0.4 mm. In this embodiment, the threshold value is set to 0.15 mm.

[0097] If the position misalignment amount is less than the threshold value (the step **S104**: “NO”), the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small). Therefore, the controller CU ends the misalignment identification and correction processing (the step **S4** in FIG. 6), and proceeds to the step **S5**. Meanwhile, if the position misalignment amount is the threshold value or more (the step **S104**: “YES”), it may be determined that the edge ring ER is misaligned (or the position misalignment amount is large). In this case, the controller CU detects an error of the edge ring ER, and proceeds to step **S105** in FIG. 7.

[0098] In the step **S105**, the controller CU raises the edge ring ER by the supporting pins **521** of the plasma processing apparatus **1**, and enters the transfer robot TR1 into the plasma processing space **10s** to transfer the edge ring ER. Accordingly, as illustrated in FIG. 8D, the edge ring ER is held by the transfer robot TR1 in a state in which the edge ring ER is misaligned.

[0099] After that, the controller CU performs a correction movement of correcting the misalignment of the edge ring ER by the transfer robot TR1 that holds the edge ring ER (step **S106** in FIG. 7). In a configuration that detects the misalignment of the edge ring ER by the position detection sensor S1 or S2, in the step **S103**, the position misalignment amount of the edge ring ER has already been acquired. Therefore, as illustrated in FIG. 8E, the controller CU sets a movement amount in the correction movement of the transfer robot TR1 based on the acquired position misalignment amount, and horizontally moves the transfer robot TR1 by the movement amount. In the example of FIG. 8E, the transfer robot TR1 is horizontally moved in a right direction by the movement amount. As a result, the edge ring ER is returned to a normal position above the substrate support **11**.

[0100] The controller CU receives the edge ring ER from the transfer robot TR1 by the supporting pins **521**, and places the edge ring ER on the ring support surface **112b** again by lowering the supporting pins **521** after withdrawal of the transfer robot TR1 (step **S107** in FIG. 7). Accordingly, as illustrated in FIG. 8F, the edge ring ER is supported on the ring support surface **112b** in a state in which the position misalignment is resolved. When the step **S107** is ended, the misalignment identification and correction processing (step **S4**) is ended. Further, after receiving the edge ring ER by the supporting pins **521**, the controller CU may perform an unloading processing of unloading the edge ring ER from the processing module PM1 without lowering the supporting pins **521**.

[0101] Back to FIG. 6, in the step **S5**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM1. Specifically, the controller CU raises the edge ring ER by the supporting pins **521** of the plasma processing apparatus **1** and receives the

edge ring ER via the transfer robot TR1. After placing the edge ring ER (after the step S107), electrostatic attraction by the electrostatic chuck 112 is not performed. Therefore, when raising the edge ring ER by the supporting pins 521, the edge ring ER may be disposed without being misaligned from a holding position of the transfer robot TR1. The transfer robot TR1 receives the edge ring ER in which no misalignment occurs, and unloads the edge ring ER from the processing module PM1 after the lowering of the supporting pins 521. At this time, the transfer robot TR1 may smoothly unload the edge ring ER without interfering with other parts.

[0102] In the step S6, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by transferring the edge ring ER by the transfer robot TR1. Since the transfer robot TR1 holds the edge ring ER in which no misalignment occurs, it is possible to stably transfer the edge ring ER into the ring accommodation module RSM without interfering with other parts of the ring accommodation module RSM or the like.

[0103] As described above, in the substrate processing system PS and the transfer method, a misalignment of the edge ring ER is identified when unloading the edge ring ER from the processing module PM1, and a correction is performed when the misalignment occurs. Therefore, it is possible to transfer the edge ring ER without interfering with other parts. Accordingly, it is possible to prevent the edge ring ER from dropping off the transfer robot TR1, and to avoid damage to the edge ring ER or other parts, and the like. Thus, it is possible to for the substrate processing system PS to reduce an opportunity for stopping the substrate processing system PS for performing a maintenance.

[0104] Further, the substrate processing system PS and the transfer method are not limited to the embodiment, and may have various modifications. For example, the transfer method illustrated in FIG. 6 has described a case where only the edge ring ER is exchanged, but may be applied even in a case where a cover ring CR (the outer ring 113b) is exchanged. By performing a misalignment identification and correction processing even when unloading the cover ring CR, it is possible to stably transfer the cover ring CR.

[0105] The substrate support 11 of each of the processing modules PM1 to PM7 is not limited to a configuration that electrostatically attracts the substrate W and the ring 113 by the electrostatic chuck 112. For example, the substrate support 11 may employ a mechanism that provides an attractive force to the substrate W and the ring 113, a mechanism that mechanically hangs and fixes the ring 113, or the like. In this case, since the ring 113 is adhered to the ring support surface 112b or other parts, the ring 113 may be misaligned when the supporting pins 521 are raised. Thus, it is possible for the substrate processing system PS to transfer the ring 113 while suppressing the misalignment of the ring 113 by performing the transfer method according to the embodiment.

[0106] In addition, in the misalignment identification and correction processing, after placing the edge ring ER on the ring support surface 112b in the step S102, the index related to the position misalignment amount of the edge ring ER is acquired by the position detection sensor S1 or S2 of the transfer robot TR1. However, in the misalignment identification and correction processing, the index related to the position misalignment amount of the edge ring ER may be acquired by the position detection sensor S1 or S2 of the transfer robot TR1 in a state in which the edge ring ER is supported (raised) by the supporting pins 521.

[0107] For example, as illustrated in FIG. 9, in step S111, the controller CU raises the edge ring ER by the supporting pins 521, and stops the supporting pins 521 at a timing at which the edge ring ER is separated from the ring support surface 112b. Further, the controller CU acquires a position misalignment amount by the position detection sensors S1 and S2 in a state in which the supporting pins 521 are stopped. Alternatively, the controller CU may acquire a position misalignment amount by the position detection sensors S1 and S2 at an arbitrary pin height which is higher than the position at which the edge ring ER is separated from the ring support surface 112b and at which the upper surface of the edge ring ER is below the position detection sensors S1 and S2 of the transfer robot TR1. In step S112, the controller CU detects a position misalignment amount of the edge ring

ER by the position detection sensor S1 or S2. In step S113, the controller CU determines whether the position misalignment amount of the edge ring ER is a threshold value or more. Further, if the position misalignment amount is the threshold value or more (the step S113: "YES"), the controller CU raises the edge ring ER to a transfer height by the supporting pins 521 and enters the transfer robot TR1 into the plasma processing space 10s to transfer the edge ring ER (step S114). In step S115, the controller CU performs a correction movement of correcting a misalignment of the edge ring ER by the transfer robot TR1 that holds the edge ring ER. Further, in step S116, the controller CU receives the edge ring ER from the transfer robot TR1 by the supporting pins 521, and places the edge ring ER on the ring support surface 112b again by lowering the supporting pins 521 after withdrawal of the transfer robot TR1. Accordingly, operations of lowering the edge ring ER in the step S102 in FIG. 7 and raising the edge ring ER in the step S105 in FIG. 7 may be omitted, thus more efficiently performing a processing. Further, in the step S115, the controller CU may move and correct the transfer robot TR1 in consideration of the position misalignment amount of the edge ring ER, and then receive the edge ring ER and unload the edge ring ER as it is. Accordingly, the step S116 may be omitted.

[0108] In addition, in the misalignment identification and correction processing, the misalignment of the edge ring ER has been resolved by receiving the edge ring ER by the transfer robot TR1, and then performing the correction movement. However, in the misalignment identification and correction processing, before the transfer robot TR1 receives the edge ring ER held by the supporting pins 521 in the step S105 in FIG. 7, the transfer robot TR1 may be corrected and moved in advance by the position misalignment amount, and then the edge ring ER may be transferred to the transfer robot TR1. Accordingly, for example, the step S106 or S107 in FIG. 7 may be omitted, thus much more efficiently performing a processing.

[0109] In addition, in the misalignment identification and correction processing (the step S4), the substrate processing system PS may perform a cleaning processing in the plasma processing chamber 10 by stopping at a predetermined height while raising the edge ring ER by the supporting pins 521, may perform the cleaning processing in the plasma processing chamber 10 after raising the supporting pins 521 to the transfer height, or may perform the cleaning processing in the plasma processing chamber 10 while raising the edge ring ER. Alternatively, in the unloading processing from any one of the processing modules PM1 to PM7 (the step S5), the substrate processing system PS may perform the cleaning processing in the plasma processing chamber 10 by stopping at a predetermined height while raising the edge ring ER by the supporting pins 521, may perform the cleaning processing in the plasma processing chamber 10 after raising the supporting pins 521 to the transfer height, or may perform the cleaning processing in the plasma processing chamber 10 while raising the edge ring ER. The cleaning processing may use, for example, a method such as wafer-less dry cleaning (WLDC) or a wafer with dry cleaning (WWDC) performed by placing, on the substrate support 11, a dummy wafer that has a diameter smaller than the diameter of the substrate W and equal to the diameter of the substrate support surface 112a. In the case of the WWDC, the dummy wafer having the same diameter as the substrate support surface 112a is placed on the substrate support 11 until before the steps S4 and S5.

[0110] In addition, from after the substrate processing in the step S1 to the charge removal processing in the step S3, the substrate processing system PS may perform wafer-less dry cleaning, or may perform dry cleaning by placing a dummy wafer having the same diameter as the substrate W or a dummy wafer having a diameter smaller than the diameter of the substrate W and equal to the diameter of the substrate support surface 112a. Further, before the ring for exchange is loaded after the step S5, the substrate processing system PS may perform wafer-less dry cleaning, or may perform dry cleaning by placing a dummy wafer having the same diameter as the substrate W or a dummy wafer having a diameter smaller than the diameter of the substrate W and equal to the diameter of the substrate support surface 112a.

[0111] The cleaning processing is performed in the misalignment identification and correction

processing or the unloading processing, so that it is possible to clean the edge ring ER before unloading the edge ring ER to the vacuum transfer chamber, thereby suppressing the vacuum transfer chamber from being contaminated by a deposit attached to the edge ring ER. In addition, the cleaning processing is performed at a timing at which the edge ring ER is raised, so that it is possible to remove a deposit attached to the ring support surface **112b**. In addition, the substrate processing system PS performs the cleansing processing on the edge ring ER during an unloading operation, so that it is possible to improve a throughput of the entire processing as compared with a case where the cleaning processing is separately performed. In addition, the deposit attached to the ring support surface is removed by performing the cleaning processing after the edge ring is unloaded and before the edge ring for exchange is loaded, so that it is possible to suppress an attraction failure of the edge ring for exchange.

[0112] FIG. **10** is a flowchart illustrating a transfer method according to a second example. The transfer method according to the second example performs state identification and transfer of the edge ring ER by the position detection sensor **S1** or **S2** of the transfer robot **TR1**. Further, in description of the transfer method according to the second example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module **PM1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules **PM2** to **PM7**, the same method as a case where the processing module for the exchange target is the processing module **PM1** may be used.

[0113] When performing the steps to the step **S3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps **S201** to **S210** as the transfer method according to the second example. The steps **S201** to **S210** are performed as the controller CU controls each part of the substrate processing system PS.

[0114] The steps **S201** to **S204** may be the same as the steps **S101** to **S104**. However, if a position misalignment amount of the edge ring ER is a threshold value or more in the step **S204** (the step **S204**: “YES”), the controller CU may proceed to the step **S205**. If the position misalignment amount of the edge ring ER is less than the threshold value, the controller CU may proceed to the step **S208** (the step **S204**: “NO”).

[0115] In the step **S205**, the controller CU raises the edge ring ER by the supporting pins **521**, enters the transfer robot **TR1** into the plasma processing space **10s**, and then lowers the supporting pins **521** to transfer the edge ring ER to the transfer robot **TR1**. Accordingly, the edge ring ER is held by the transfer robot **TR1** in a state in which the edge ring ER is misaligned.

[0116] After that, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module **PM1** (the step **S206**). By performing the unloading processing after the step **S205**, the transfer robot **TR1** unloads the edge ring ER from the processing module **PM1** while holding the edge ring ER in the state in which the edge ring ER is misaligned.

[0117] Further, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by transferring the edge ring ER by the transfer robot **TR1** (the step **S207**). In the loading processing, when loading the edge ring ER into an accommodation module (e.g., a cassette which is not illustrated) of the ring accommodation module RSM, the controller CU sets a movement amount of the transfer robot **TR1**, based on the position misalignment amount acquired in the step **S203**, and corrects and moves the transfer robot **TR1**. Accordingly, the edge ring ER is returned to the normal position by the transfer robot **TR1** without interfering with the ring accommodation module RSM. Thus, the controller CU may smoothly accommodate the edge ring ER without interfering with other parts.

[0118] Meanwhile, if the position misalignment amount is less than the threshold value, the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small), and hence the controller CU performs ordinary transfer (of not performing movement and correction) in the steps **S208** to **S210**.

[0119] Specifically, the controller CU raises the edge ring ER by the supporting pins **521** of the

plasma processing apparatus **1** and enters the transfer robot **TR1** into the plasma processing space **10s** to transfer the edge ring **ER** (the step **S208**). Further, the controller **CU** performs an unloading processing of unloading the edge ring **ER** from the processing module **PM1** (the step **S209**). Further, the controller **CU** transfers the edge ring **ER** by the transfer robot **TR1**, and loads the edge ring **ER** into the ring accommodation module **RSM** (the step **S210**). When loading the edge ring **ER** into the ring accommodation module **RSM**, the transfer robot **TR1** may accommodate the edge ring **ER** in the accommodation module without moving and correcting the edge ring **ER**. That is, the unloading processings in the steps **S206** and **S209** correspond to the unloading processing in the step **S5** in FIG. **6**, and the loading processings in the steps **S207** and **S210** correspond to the loading processing in the step **S6** in FIG. **6**.

[0120] As such, in a case where the edge ring **ER** is misaligned, the transfer method according to the second example performs movement and correction when the edge ring **ER** is loaded into the ring accommodation module **RSM**, thus suppressing interference between the edge ring **ER** and other parts. Further, the transfer method according to the second example may also have various modifications. For example, in the misalignment identification and correction processing, an index related to a position misalignment amount of the edge ring **ER** may be acquired by the position detection sensor **S1** or **S2** of the transfer robot **TR1** in a state in which the edge ring **ER** is supported by the supporting pins **521**. That is, in the step **S201**, the controller **CU** may stop the supporting pins **521** at a position at which the edge ring **ER** is separated from the ring support surface **112b** and then acquire a position misalignment amount by the position detection sensor **S1** or **S2**, or may acquire a position misalignment amount by the position detection sensors **S1** and **S2**, at an arbitrary pin height which is higher than the position at which the edge ring **ER** is separated from the ring support surface **112b** and at which the upper surface of the edge ring **ER** is below the position detection sensors **S1** and **S2** (see FIG. **9**).

[0121] Further, the controller **CU** may select implementation of the transfer method according to the first example or implementation of the transfer method according to the second example based on the position misalignment amount of the edge ring **ER**. For example, the controller **CU** has a threshold value for selection of a transfer method, and may determine the transfer method according to the first example if the position misalignment amount is equal to or larger than the threshold value for selection, or may determine the transfer method according to the second example if the position misalignment amount is less than the threshold value for selection. The threshold value for selection is a value larger than a threshold value for determining movement and correction. Accordingly, when the position misalignment amount of the edge ring **ER** is large, a misalignment of the edge ring **ER** is corrected in the processing module **PM1**, so that more stable transfer is possible. Meanwhile, when the position misalignment amount of the edge ring **ER** is small even though movement and correction is necessary, a misalignment of the edge ring **ER** is corrected when the edge ring **ER** is transferred to the ring accommodation module **RSM**, so that it is possible to improve transfer efficiency.

[0122] The substrate processing system **PS** is not limited to a configuration of loading the ring **113** (the edge ring **ER**) in any one of the processing modules **PM1** to **PM7** directly into the ring accommodation module **RSM** by the transfer robot **TR1**. The substrate processing system **PS** may transfer an upper ring **222** of a ring **220** to a standby transfer module **LM**, for example, when a plasma processing apparatus **1A** shown in FIG. **11** is applied. FIG. **11** is a schematic cross-sectional view illustrating another example of the plasma processing apparatus.

[0123] The plasma processing apparatus **1A** shown in FIG. **11** is different from the plasma processing apparatus **1** in that the plasma processing apparatus **1A** includes a substrate support **16** instead of the substrate support **11** and includes a ring assembly **220** (ring **220**) instead of the ring assembly **113**. Other parts may be the same as the plasma processing apparatus **1**. Hereinafter, differences from the plasma processing apparatus **1** are mainly described.

[0124] The plasma processing apparatus **1A** includes the substrate support **16**. The substrate

support **16** is installed inside the plasma processing chamber **10**. The substrate support **16** supports a substrate **W**. The substrate support **16** is supported by a support **17**. The support **17** extends upward from a bottom portion of the plasma processing chamber **10**. The support **17** has a cylindrical shape. The support **17** is made of an insulating material such as quartz.

[0125] The substrate support **16** has a first region **161** and a second region **162**. The first region **161** supports the substrate **W**. The first region **161** is a substantially circular region in a plan view. The first region **161** may include a base **18** and an electrostatic chuck **19**. The first region **161** may be configured by a portion of the base **18** and a portion of the electrostatic chuck **19**. The base **18** and the electrostatic chuck **19** are installed inside the plasma processing chamber **10**. The base **18** is made of a conductive material such as aluminum. The base **18** has a substantially disk shape. The base **18** constitutes a lower electrode.

[0126] The substrate support **16** includes a main body **2** and the ring assembly **220**. The main body **2** includes the base **18** and the electrostatic chuck **19**. The main body **2** has a substrate support region **2a** for supporting the substrate **W**, an annular region **2b** for supporting the ring assembly **220**, and a sidewall **2c** that extends vertically between the substrate support region **2a** and the annular region **2b**. The annular region **2b** surrounds the substrate support region **2a**. The annular region **2b** is located below the substrate support region **2a**. Therefore, an upper end of the sidewall **2c** is connected to the substrate support region **2a**, and a lower end of the sidewall **2c** is connected to the annular region **2b**.

[0127] A flow path **18f** is formed in the base **18**. The flow path **18f** is a flow path through which a heat exchange medium flows. As the heat exchange medium, a liquid refrigerant or a refrigerant (e.g., chlorofluorocarbon) that cools the base **18** by vaporization of the liquid refrigerant is used. A supply device of the heat exchange medium (e.g., a chiller unit) is connected to the flow path **18f**. The supply device is installed outside the plasma processing chamber **10**. The heat exchange medium is supplied to the flow path **18f** from the supply device. The heat exchange medium supplied to the flow path **18f** is returned to the supply device.

[0128] The electrostatic chuck **19** is installed on the base **18**. The substrate **W** is placed on the first region **161** and on the electrostatic chuck **19** when processed in the plasma processing chamber **10**.

[0129] The second region **162** extends radially outward from the first region **161** to surround the first region **161**. The second region **162** is a substantially annular region in a plan view. The ring assembly **220** is placed on the second region **162**. The second region **162** may include the base **18**. The second region **162** may include the electrostatic chuck **19**. The second region **162** may be composed of another portion of the base **18** and another portion of the electrostatic chuck **19**. The substrate **W** is placed in a region surrounded by the ring assembly **220** and on the electrostatic chuck **19**. Details of the ring assembly **220** will be described later.

[0130] A through-hole **162h** is formed in the second region **162**. The main body **2** has the through-hole **162h** formed between the annular region **2b** and a lower surface **2d** of the main body **2**. The through-hole **162h** is formed in the second region **162** to extend along a vertical direction. A plurality of through-holes **162h** are formed in the second region **162**. A number of the through-holes **162h** may be equal to a number of lift pins **72** of a lift mechanism **70** which will be described later. Each through-hole **162h** is disposed parallel with a corresponding lift pin **72** on a straight line.

[0131] The electrostatic chuck **19** includes a main body **19m** and an electrode **19e**. The main body **19m** is made of a dielectric such as aluminum oxide or aluminum nitride. The main body **19m** has a substantially disk shape. The electrode **19e** is installed in the main body **19m**. The electrode **19e** has a film shape. A direct current power source is electrically connected to the electrode **19e** via a switch. When a voltage from the direct current power source is applied to the electrode **19e**, an electrostatic attraction force is generated between the electrostatic chuck **19** and the substrate **W**. By the generated electrostatic attraction force, the substrate **W** is attracted to the electrostatic chuck **19** and held by the electrostatic chuck **19**.

[0132] The plasma processing apparatus **1A** further includes an outer peripheral member **27**. The

outer peripheral member **27** extends circumferentially and radially outward from the substrate support **16** to surround the substrate support **16**. The outer peripheral member **27** may extend circumferentially and radially outward from the support **17** to surround the support **17**. The outer peripheral member **27** may be composed of one or more parts. The outer peripheral member **27** may be formed of an insulating material such as quartz.

[0133] The ring assembly **220** and the substrate support **16** will now be described in more detail. The ring assembly **220** includes a lower ring **221** and an upper ring **222**.

[0134] Each of the lower ring **221** and the upper ring **222** has an annular shape. Each of the lower ring **221** and the upper ring **222** is formed of a material appropriately selected according to a plasma processing performed in the plasma processing apparatus **1A**. Each of the lower ring **221** and the upper ring **222** is formed of, for example, silicon or silicon carbide.

[0135] The lower ring **221** is disposed on the annular region **2b**. The lower ring **221** may be placed on the second region **162** and on the electrostatic chuck **19**. The lower ring **221** may be placed on a part other than the electrostatic chuck **19** in the second region **162**.

[0136] A lower surface of the upper ring **222** is substantially flat. The lower surface of the upper ring **222** includes a tapered surface and defines a recess. The lower surface of the upper ring **222** defines a plurality of recesses. A number of tapered surfaces of the upper ring **222** and a number of recesses may be the same as the number of lift pins **72** of the lift mechanism **70**. Each recess has a size in which a tip end of a second rod portion **722** of a corresponding lift pin **72** is fitted into the recess. The upper ring **222** is disposed on the lower ring **221** such that each recess is aligned with corresponding lift pins **72** and corresponding through-holes **221h** on a straight line.

[0137] The upper ring **222** is accommodated in a recess of the lower ring **221**. The lower ring **221** and the upper ring **222** are configured such that when disposed on the annular region **2b**, an upper surface of an outer portion of the lower ring **221** and an upper surface of the upper ring **222** are substantially at the same height as an upper surface of the substrate **W** on the substrate support region **2a**. The upper ring **222** includes an inner peripheral surface **222a** that faces an end surface of the substrate **W** on the substrate support region **2a** when the lower ring **221** and the upper ring **222** are disposed on the annular region **2b**.

[0138] The substrate support **16** includes the lift mechanism **70**. The lift mechanism **70** includes the lift pins **72**, and is configured to raise and lower the lower ring **221** and the upper ring **222**. The lift mechanism **70** includes a plurality of lift pins **72**. The number of lift pins **72** may be any number as long as the lift pins **72** are capable of supporting, raising and lowering the ring assembly **220**. The number of lift pins **72** may be, for example three.

[0139] Each lift pin **72** may be formed of an insulating material. Each lift pin **72** may be formed of, for example, sapphire, alumina, quartz, silicon nitride, aluminum nitride, or resin. Each lift pin **72** includes a first rod portion **721** and the second rod portion **722**. The first rod portion **721** extends in the vertical direction. The first rod portion **721** includes a first upper end surface **721t**. The first upper end surface **721t** may come in contact with a lower surface of the lower ring **221**.

[0140] The second rod portion **722** extends vertically upward from the first rod portion **721**. The second rod portion **722** is narrowed with respect to the first rod portion **721** to expose the first upper end surface **721t**. Each of the first rod portion **721** and the second rod portion **722** has a columnar shape. A diameter of the first rod portion **721** is larger than a diameter of the second rod portion **722**. The second rod portion **722** is capable of moving up and down through the through-hole **221h**. A length of the second rod portion **722** in the vertical direction is longer than a vertical thickness of a region of the lower ring **221** on which the upper ring **222** is placed.

[0141] The second rod portion **722** has a second upper end surface **722t**. The second upper end surface **722t** may come in contact with the upper ring **222**. The tip end of the second rod portion **722** including the second upper end surface **722t** may be formed in a tapered shape to be fitted into a corresponding recess in the upper ring **222**.

[0142] The second rod portion **722** may include a first portion **722a** and a second portion **722b**. The

first portion **722a** has a columnar shape and extends upward from the first rod portion **721**. The second portion **722b** has a columnar shape and extends upward from the first portion **722a**. The second portion **722b** includes the second upper end surface **722t**. A width of the first portion **722a** is larger than a width of the second portion **722b**.

[0143] The first rod portion **721**, the first portion **722a**, and the second portion **722b** may have a columnar shape. A diameter of the first rod portion **721** is larger than a diameter of the first portion **722a**, and the diameter of the first portion **722a** is larger than a diameter of the second portion **722b**.

[0144] The second rod portion **722** may include a third portion **722c**. The third portion **722c** extends between the first portion **722a** and the second portion **722b**. The third portion **722c** has a tapered surface.

[0145] The lift mechanism **70** includes one or more drives **74**. The one or more drives **74** are configured to raise and lower the plurality of lift pins **72**. Each of the one or more drives **74** may include, for example, a motor.

[0146] For example, the upper ring **222** is stored in a ring accommodation container **CS2** loaded into the load port **LP4** of the atmospheric transfer module **LM**. The ring accommodation container **CS2** corresponds to a ring accommodation module. The controller **CU** unloads the upper ring **222** of any one of the processing modules **PM1** to **PM7** to the vacuum transfer module **TM**, and transfers the upper ring **222** to any one of the load lock modules **LL1** to **LL3**. Further, when unloading the upper ring **222** from any one of the load lock modules **LL1** to **LL3** using the transfer robot **TR2** of the atmospheric transfer module **LM**, the controller **CU** loads and accommodate the upper ring **222** in the ring accommodation container **CS2**.

[0147] Correction and movement when a position misalignment occurs in the upper ring **222** may be performed in any one of the processing modules **PM1** to **PM7**, or may be performed when the transfer robot **TR1** loads and places the upper ring **222** in any one of the load lock modules **LL1** to **LL3**. Alternatively, the correction and movement may be performed when the transfer robot **TR2** of the atmospheric transfer module **LM** loads the upper ring **222** into the ring accommodation container **CS2**. Further, in the transfer method, the same processing as above may be performed even when transferring the lower ring **221** in addition to the upper ring **222**. In addition, for example, the processing modules **PM1** to **PM7** of the substrate processing system **PS** are not limited to the ring assemblies **113** and **220** including a combination of a plurality of members, and may be configured to use a ring including one member. In this case, the transfer methods according to the first and second examples may be applied, and transfer methods according to third to fifth examples, which will be described later, may be applied.

[0148] Next, a transfer method when performing state identification and transfer of the edge ring **ER** by the leakage check illustrated FIG. 5B is described. FIG. 12 is a flowchart illustrating a processing flow in a transfer method according to a third example. Further, in description of the transfer method according to the third example, a case where the ring **113** is transferred between the ring accommodation module **RSM** and the processing module **PM1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules **PM2** to **PM7**, the same method as a case where the processing module for the exchange target is the processing module **PM1** may be used.

[0149] When performing the steps to the step **S3** in the operation sequence in FIG. 6, the controller **CU** of the substrate processing system **PS** performs steps **S301** to **S308** as the transfer method according to the third example. The steps **S301** to **S308** are performed as the controller **CU** controls each part of the substrate processing system **PS**.

[0150] The steps **S301** and **S302** may be the same as the steps **S101** and **S102**. In addition, in order to perform a leakage check after the step **S302** (before the step **S303**), the controller **CU** performs a processing of electrostatically attracting the edge ring **ER** to the ring support surface **112b**.

[0151] In the step **S303**, in order to acquire an index related to a position misalignment amount of

the edge ring ER, the controller CU performs a leakage check by the plasma processing apparatus **1**. As illustrated in FIG. 5B, in the leakage check, the controller **90** of the plasma processing apparatus **1** supplies a gas to a gap between the ring support surface **112b** to which the edge ring ER is electrostatically attracted and the edge ring ER, and stops the supply of the gas when a pressure in the gap reaches a target pressure. Further, the controller **90** measures a pressure in a flow path after a predetermined time elapses from the stop of the supply by the pressure sensor **67**. [0152] The controller CU acquires, as the index related to the position misalignment amount of the edge ring ER, a measurement result of the pressure sensor **67** after the predetermined time elapses, and determines a position misalignment of the edge ring ER (the step S304 in FIG. 12). For example, if the pressure is a pre-stored pressure threshold value or more, the controller CU determines that the edge ring ER is not misaligned (normal). Meanwhile, if the pressure less than the pressure threshold value, the controller CU determines that the edge ring ER is misaligned (abnormal). Further, if the edge ring ER is not misaligned (the step S304: “YES”), the controller CU proceeds to the step S305. If the edge ring ER is misaligned (the step S304: “NO”), the controller CU proceeds to the step S308.

[0153] In the step S305, the controller CU ends the leakage check and the misalignment identification and correction processing. Further, in the step S306, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM1 by the transfer robot TR1. The step S306 corresponds to the step S5 in FIG. 6. Further, since the leakage check is performed in a state in which the edge ring ER is electrostatically attracted, the controller CU may perform a charge removal processing of the edge ring ER before the step S306 (or the step S308 to be described later). Further, in the step S307, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR1. The step S307 corresponds to the step S6 in FIG. 6.

[0154] Meanwhile, when a misalignment occurs in the edge ring ER, in the step S308, the controller CU performs a misalignment resolution operation for resolving the misalignment of the edge ring ER. In the misalignment resolution operation, for example, as illustrated in FIGS. 13A to 13C, a recess **114** formed in the edge ring ER and a configuration of each supporting pin **521** may be used.

[0155] Specifically, in a lower surface of the edge ring ER, the recess **114** of the edge ring ER is provided as many as the supporting pins **521**. As illustrated in FIG. 13A, each recess **114** includes a flat bottom portion **114a** and a tapered portion **114b** having an inner diameter widened toward an opening portion while surrounding a periphery of the bottom portion **114a**. The bottom portion **114a** is formed wider than the outer diameter of the upper pin **524** of the supporting pin **521**, and have, for example, about 2 mm. The tapered portion **114b** is smoothly connected to the bottom portion **114a** and the lower surface of the edge ring ER through rounded edge portions.

[0156] When a position misalignment occurs on the edge ring ER having the recess **114**, for example, the upper end surface **524a** of the supporting pin **521** and the tapered portion **114b** face each other. In this case, the controller CU performs, as the misalignment resolution operation, an operation of raising the supporting pins **521** by the plasma processing apparatus **1**.

[0157] That is, as illustrated in FIG. 13B, when raising the supporting pin **521**, the upper end surface **524a** of the supporting pin **521** comes in contact with the tapered portion **114b** of the edge ring ER. Therefore, the tapered portion **114b** slides along a slope of the tapered portion **114b** with respect to the supporting pin **521**. As a result, as illustrated in FIG. 13C, the edge ring ER moves in a horizontal direction to guide the supporting pin **521** to a substantially central portion of the bottom portion **114a**. By the movement in the horizontal direction, the misalignment of the edge ring ER is resolved.

[0158] Back to FIG. 12, when performing the misalignment resolution operation in the step S308, the controller CU returns to the step S303 to perform the leakage check again. This is for the purpose of identifying whether the misalignment of the edge ring ER has been resolved by the

misalignment resolution operation. If the measured pressure is less than the pressure threshold value, the misalignment of the edge ring ER is not resolved, and hence the controller CU repeats the misalignment resolution operation (the step **S308**). Meanwhile, if the measured pressure is the pressure threshold value or more, the controller CU proceeds to the step **S305** to end the misalignment identification and correction processing.

[0159] As described above, in the substrate processing system PS and the transfer method, it is possible to identify the misalignment of the edge ring ER by performing the leakage check. Further, in the substrate processing system PS and the transfer method, it is possible to correct the misalignment of the edge ring ER by using the recess **114** of the edge ring ER and the shape of the supporting pin **521**. As a result, the transfer robot TR1 may transfer the edge ring ER without interfering with other parts.

[0160] FIG. **14** is a flowchart illustrating a transfer method according to a fourth example. The transfer method according to the fourth example performs state identification and transfer of the edge ring ER by performing a leakage check in the processing module PM1. Further, in description of the transfer method according to the fourth example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module PM1 is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM2 to PM7, the same method as a case where the processing module for the exchange target is the processing module PM1 may be used.

[0161] When performing the steps to the step **S3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps **S401** to **S410** as the transfer method according to the fourth example. The steps **S401** to **S410** are performed as the controller CU controls each part of the substrate processing system PS.

[0162] The steps **S401** to **S404** may be the same as the steps **S301** to **S304**. In addition, in order to perform a leakage check after the step **S402** (before the step **S403**), the controller CU performs a processing of electrostatically attracting the edge ring ER to the ring support surface **112b**. Further, if the pressure of the pressure sensor **67** is less than the pressure threshold value in the step **S404** (the step **S404**: “NO”), the controller CU proceeds to the step **S405**. If the pressure of the pressure sensor **67** is the pressure threshold value or more (the step **S404**: “YES”), the controller CU omits the steps **S405** to **S408** and proceeds to the step **S409**.

[0163] In the step **S405**, the controller CU operates the transfer robot TR1 and detects a position (an index related to a position misalignment amount) of the edge ring ER by the position detection sensor **S1** or **S2**. Accordingly, the controller CU may obtain a gap amount (a position misalignment amount) between the substrate support surface **112a** and the edge ring ER by receiving a detection result of the position detection sensor **S1** or **S2**.

[0164] Further, controller CU may perform correction and movement illustrated in of FIGS. **8D** to **8F** by utilizing the position misalignment amount. Further, since the leakage check is performed in a state in which the edge ring ER is electrostatically attracted, the controller CU may perform a charge removal processing of the edge ring ER from the determination “YES” in the step **S404** and before the step **S406**. Therefore, in the step **S406**, the controller CU raises the edge ring ER by the supporting pins **521** in a state in which the attraction of the edge ring ER to the ring support surface **112b** is resolved. That is, after the position misalignment amount is detected, it is possible to transfer the edge ring ER to the transfer robot TR1 while the position of the edge ring ER is prevented from being further misaligned. An operation in the step **S407** may be the same operation as the step **S106** in FIG. **7**, and an operation in the step **S408** may be the same operation as the step **S107** in FIG. **7**.

[0165] Further, in the step **S409**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM1 by the transfer robot TR1. The step **S409** corresponds to the step **S5** in FIG. **6**. Further, in the step **S410**, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the

transfer robot TR1. The step S410 corresponds to the step S6 in FIG. 6.

[0166] As described above, when recognizing a misalignment of the edge ring ER in the leakage check, the transfer method according to the fourth example detects a position of the edge ring ER by the position detection sensor S1 or S2 of the transfer robot S1 or S2. Accordingly, the controller CU may certainly detect a position misalignment amount of the edge ring ER, and thus it is possible to correct and move the edge ring ER with a high degree of precision. As a result, the transfer method may transfer the edge ring ER without interfering with other parts.

[0167] FIG. 15 is a flowchart illustrating a transfer method according to a fifth example. The transfer method according to the fifth example performs state identification and transfer of the edge ring ER by performing a leakage check in the processing module PM1. Further, in description of the transfer method according to the fifth example, a case where the ring 113 is transferred between the ring accommodation module RSM and the processing module PM1 is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM2 to PM7, the same method as a case where the processing module for the exchange target is the processing module PM1 may be used.

[0168] When performing the steps to the step S3 in the operation sequence in FIG. 6, the controller CU of the substrate processing system PS performs steps S501 to S511 as the transfer method according to the fifth example. The steps S501 to S511 are performed as the controller CU controls each part of the substrate processing system PS.

[0169] The steps S501 to S504 may be the same as the steps S301 to S304. In addition, in order to perform a leakage check after the step S502 (before the step S503), the controller CU performs a processing of electrostatically attracting the edge ring ER to the ring support surface 112b. Further, if the pressure of the pressure sensor 67 is less than the pressure threshold value in the step S504 (the step S504: "NO"), the controller CU proceeds to the step S505. If the pressure of the pressure sensor 67 is the pressure threshold value or more, the controller CU proceeds to the step S509 (the step S504: "YES").

[0170] The steps S505 and S506 may be the same as the steps S205 and S206 in the transfer method according to the second example. Further, since the leakage check is performed in a state in which the edge ring ER is electrostatically attracted, the controller CU may perform a charge removal processing of the edge ring ER before the step S505 (or before the step S509 to be described later).

[0171] In the step S507, the controller CU detects a position misalignment amount of the edge ring ER by the position detection sensors S11 and S12 installed in the vicinity of the gate valve (not illustrated) which partitions the vacuum transfer module TM and the processing module PM1. For example, when transferring the edge ring ER by the transfer robot TR1, the controller CU performs detection by the position detection sensors S11 and S12 and calculates a position of the edge ring ER based on a position at which the position detection sensors S11 and S12 are shaded by the edge ring ER and a time for which the position detection sensors S11 and S12 are shaded by the edge ring ER. Further, based on the position of the edge ring ER and a preset reference position, the controller CU calculates a position misalignment amount (an index related to the position misalignment amount) of the edge ring ER from the reference position.

[0172] After calculating the position misalignment amount when unloading the edge ring ER, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR1 (the step S508). In the loading processing, when loading the edge ring ER into the accommodation module (e.g., the cassette which is not illustrated) of the ring accommodation module RSM, the controller CU sets a movement amount of the transfer robot TR1 based on the position misalignment amount acquired in the step S507, and corrects and moves the transfer robot TR1. Accordingly, the edge ring ER is returned to the normal position by the transfer robot TR1 without interfering with the ring accommodation module RSM. Thus, the controller CU may smoothly accommodate the edge ring ER without interfering with

other parts.

[0173] Meanwhile, if a leakage amount is less than a threshold value, the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small), and hence the controller CU performs ordinary transfer (of not performing movement and correction) in the steps **S509** to **S511**. The steps **S509** to **S511** may be the same as the steps **S208** to **S210**.

[0174] As described above, when recognizing a misalignment of the edge ring ER in the leakage check, the transfer method according to the fifth example detects the position of the edge ring ER by the position detection sensors **S11** and **S12** of the vacuum transfer module TM. In this case, the controller CU may certainly detect a position misalignment amount of the edge ring ER, and thus it is possible to correct and move the edge ring ER with a high degree of precision when loading the edge ring ER into the ring accommodation module RSM.

[0175] In addition, when detecting a position of the edge ring ER by the position detection sensor **S11** and **S12** of the vacuum transfer module TM, movement and correction may be performed in the processing module PM1 by returning the edge ring ER to the processing module PM1 for the time being without transferring the edge ring ER to the ring accommodation module RSM after the detection.

[0176] Next, a transfer method when performing state identification and transfer of the edge ring ER by the camera CM of the transfer robot TR1 illustrated in FIG. 5C, is described. FIG. 16 is a flowchart illustrating a processing flow in a transfer method according to a sixth example. Further, in description of the transfer method according to the sixth example, a case where the ring 113 is transferred between the ring accommodation module RSM and the processing module PM1 is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM2 to PM7, the same method as a case where the processing module for the exchange target is the processing module PM1 may be used.

[0177] When performing the steps to the step **S3** in the operation sequence in FIG. 6, the controller CU of the substrate processing system PS performs steps **S601** to **S609** as the transfer method according to the sixth example. The steps **S601** to **S609** are performed as the controller CU controls each part of the substrate processing system PS.

[0178] The steps **S601** and **S602** may be the same as the steps **S101** and **S102** in the transfer method according to the first example.

[0179] In the step **S603**, the controller CU acquires image information as an index related to a position misalignment amount by operating the transfer robot TR1 and imaging the substrate support surface 112a and the edge ring ER by the camera CM. The controller CU performs an image processing on the image information to extract a gap amount (a position misalignment amount) between the substrate support surface 112a and the edge ring ER from color or contrast information.

[0180] Therefore, based on the position misalignment amount extracted from the image information and a pre-stored threshold value, the controller CU determines whether the position misalignment amount of the edge ring ER is the threshold value or more (the step **S604**). If the position misalignment amount is less than the threshold value (the step **S604**: “NO”), the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small). Therefore, the controller CU skips the steps **S605** to **S607** and immediately proceeds to the step **S608**. Meanwhile, if the position misalignment amount is the threshold value or more (the step **S604**: “YES”), the edge ring ER may be misaligned (or the position misalignment amount may be large). In this case, the controller CU detects an error of the edge ring ER, and proceeds to step **S605**.

[0181] In addition, the steps **S605** to step **S607** may be the same as the steps **S105** to **S107** in the transfer method according to the first example.

[0182] Further, in the step **S608**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM1 by the transfer robot TR1. The step **S608**

corresponds to the step S5 in FIG. 6. Further, in the step S609, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR1. The step S609 corresponds to the step S6 in FIG. 6.

[0183] As described above, in the transfer method according to the sixth example, it is possible to obtain the position misalignment amount of the edge ring ER by imaging the edge ring ER by the camera of the transfer robot TR1. Thus, the controller CU operates the transfer robot TR1, based on the position misalignment amount of the image information, so that it is possible to transfer the edge ring ER without interfering with other parts.

[0184] In addition, a transfer method in a configuration in which the camera CM is applied to the transfer robot TR1 is not limited to the transfer method according to the sixth example, and may be used, for example, in combination with the leakage check or in combination with the position detection sensor S11 and S12 of the transfer module TM. As for the position misalignment amount of the edge ring ER, image information of the camera CM (or a detection result of the position detection sensor S1 or S2) and a detection result of the position detection sensors S11 and S12 are used, so that it is possible to further increase precision of detection or precision of correction and movement. In addition, in the transfer method according to the sixth example, when acquiring an index related to a position misalignment amount of the edge ring ER by the position detection sensor S1 or S2 of the transfer robot TR1, detection may be performed in a state in which the edge ring ER is supported by the supporting pins 521. That is, in the step S601, the controller CU may stop the supporting pins 521 at a position at which the edge ring ER is separated from the ring support surface 112b and acquire the position misalignment amount by the position detection sensor S1 or S2, or the controller CU may acquire the position misalignment amount by the position detection sensor S1 or S2 at an arbitrary pin height which is higher than the position at which the edge ring ER is separated from the ring support surface 112b and at which the upper surface of the edge ring ER is below the position detection sensors S1 and S2 (see FIG. 9).

[0185] In addition, in the configuration in which the camera CM is applied to the transfer robot TR1, as illustrated in FIG. 17, an imaging direction of the camera CM is appropriately adjusted, so that it is possible not only to image the substrate support surface 112a or the edge ring ER below the camera CM but also to image a front in a transfer direction. Therefore, it is possible to image the edge ring ER being supported by the supporting pins 521 and obliquely inclined while getting out of some supporting pins 521 using the camera CM. Further, it is possible to image the edge ring ER is supported on the electrostatic chuck 112 while getting out of all the supporting pins 521 using the camera CM.

[0186] The controller CU image-processes and analyzes such image information to recognize that the edge ring ER is inclined or that the edge ring ER gets out of all the supporting pins. For example, the camera CM images a plurality of image information in movement of the transfer robot TR1, and the controller CU performs calculation by matching positions of the edge ring ER and the transfer robot TR1 in the plurality of image information. Accordingly, the controller CU may calculate a three-dimensional form (posture and position) of the edge ring ER with a high degree of precision. Further, the controller CU may accumulate image information acquired for each image of the edge ring ER and learn recognition of the edge ring ER from images of the edge ring ER, which are included the plurality of accumulated image information. Accordingly, when acquiring image information next time, it is possible to increase extraction precision of the edge ring ER from the image information. When recognizing that the edge ring ER is inclined or that the edge ring ER gets out of all the supporting pins, the controller CU may be configured to issue an alarm.

[0187] FIG. 18 is a view illustrating a modification of FIG. 5B, which is a configuration of performing a leakage check of gas as an index related to a position misalignment amount of the edge ring. As illustrated in FIG. 18, the configuration of performing the leakage check may be a configuration including a groove 61ag that communicates with the gas supply port 61a while being recessed downward in the ring support surface 112b of the peripheral edge of the electrostatic

chuck **112**. The groove **61ag** is formed in an annular shape that goes around in a circumferential direction of the ring support surface **112b**. According to the configuration described above, the groove **61ag** may supply a gas such as He gas to the entire rear surface of the inner ring **113a** in the circumferential direction to perform a leakage check of the gas.

[0188] FIG. **19** is a flowchart illustrating an operation sequence of transfer of the edge ring ER according to a modification. In the modification shown in FIG. **19**, the substrate processing system PS and the transfer method may detect a misalignment amount of the edge ring ER by the position detection sensors **S11** and **S12** of the vacuum transfer module TM when unloading the edge ring ER from the processing module PM1. Further, the substrate processing system PS loads the edge ring ER into the ring accommodation module RSM while correcting a position of the edge ring ER based on the misalignment amount of the edge ring ER, which is detected by the position detection sensors **S11** and **S12**. In addition, even in a case where a processing module for an exchange target is any one of the processing modules PM2 to PM7, the same method as a case where the processing module for the exchange target is the processing module PM1 may be used. In this case, a position detection sensor located at a position adjacent to each processing module may be used.

[0189] The detection of the misalignment amount of the edge ring ER by the position detection sensors **S11** and **S12** may be executed by a method shown in FIGS. **20A** and **20B**. FIG. **20A** is a view illustrating a relationship between a position of the edge ring ER and positions of the position detection sensors **S11** and **S12**. FIG. **20B** is a view illustrating a change in sensor output of the position detection sensors **S11** and **S12** when transferring the edge ring ER from a position P21 to a position P24. Further, in FIG. **20B**, a time at the position P21 represents t21, a time at the position P22 represents t22, a time at the position P23 represents t23, and a time at the position P24 represents t24. Based on a position of the edge ring ER, which is detected by the position detection sensors **S11** and **S12**, and a preset reference position, the controller CU calculates a misalignment amount of the edge ring ER from the reference position. Then, the controller CU loads the edge ring ER into the ring accommodation module RSM by the transfer robot TR1 to correct the calculated misalignment amount. Accordingly, even if a position of the edge ring ER held by the upper fork FK1 or the lower fork FK2 is misaligned from the reference position, it is possible to load the edge ring ER at a predetermined position of the ring accommodation module RSM.

[0190] The position of the edge ring ER held by the upper fork FK1 or the lower fork FK2 may be calculated based on an output change of the position detection sensors **S11** and **S12**, which is generated as an inner peripheral edge passes through the position detection sensors **S11** and **S12**. For example, as illustrated in FIG. **20A**, when transferring the edge ring ER from the position P24 to the position P24, the position of the edge ring ER may be calculated based on a time T2 for which the edge ring ER moves from the position P22 to the position P23. The position P22 is a position at which the sensor output of the position detection sensor **S11** and **S12** is changed from a low (L) level to a high (H) level, and the position P23 is a position at which the sensor output of the position detection sensors **S11** and **S12** is changed from the high (H) level to the low (L) level. Specifically, as illustrated in FIG. **20B**, the position of the edge ring ER may be calculated by $T2=t23-t22$ using the time t22 at the position P22 and the time t23 at the position P23. Further, in FIGS. **20A** and **20B**, a case where a position at which the position detection sensor **S11** is shaded by the edge ring ER and a position at which the position detection sensor **S2** is shaded by the edge ring ER are the same is shown, but the positions may be different from each other.

[0191] The controller CU controls steps S701 to S706 in FIG. **19**. First, the controller CU raises the supporting pins **521** of the plasma processing apparatus **1**, and allows the edge ring ER to be spaced apart from the ring support surface **112b** of the substrate support **11** (the step S701 in FIG. **19**). When residual attraction exists between the ring support surface **112b** and the edge ring ER, a misalignment may occur in the edge ring ER when raised.

[0192] Subsequently, the controller CU enters the transfer robot TR1 below the edge ring ER in the plasma processing space **10s**, and transfers the edge ring ER to the transfer robot TR1 by lowering

the supporting pin **521** (the step **S702** in FIG. **19**).

[0193] After that, the controller CU unloads the transfer robot **TR1** from the processing module **PM1**, and detects a position of the edge ring **ER** held by the transfer robot **TR1** by the position detection sensors **S11** and **S12** (the step **S703**). The transfer robot **TR1** moves to pass through a middle of the position detection sensors **S11** and **S12**, and the center of the edge ring **ER** passes the middle when the edge ring **ER** is not misaligned. Meanwhile, when the edge ring **ER** is misaligned, a misalignment of the edge ring **ER** (a position misalignment amount and a position misalignment direction of the center of the edge ring **ER** with respect to a reference position of the transfer robot **TR**) may be calculated by the position detection sensors **S11** and **S12**.

[0194] Subsequently, the controller CU compares the position misalignment amount of the edge ring **ER** with a pre-stored threshold value, and determines whether the position misalignment amount of the edge ring **ER** is the threshold value or more (the step **S704**). If the position misalignment amount is less than the threshold value (the step **S704**: “NO”), the edge ring **ER** may not be misaligned (or the position misalignment amount may be sufficiently small). In this case, the controller CU proceeds to the step **S705**.

[0195] In the step **S705**, the controller CU loads the edge ring **ER** into the ring accommodation module **RSM** by transferring the edge ring **ER** without correcting movement of the transfer robot **TR1**. Accordingly, the substrate processing system **PS** may smoothly load the edge ring **ER** into the ring accommodation module **RSM**.

[0196] Meanwhile, if the position misalignment amount is the threshold value or more (the step **S704**: “YES”), the edge ring **ER** may be misaligned (or the position misalignment amount may be large). In this case, the controller CU proceeds to the step **S706** in FIG. **19** to load the edge ring **ER** into the ring accommodation module **RSM** while correcting the movement of the transfer robot **TR1** based on the misalignment of the edge ring **ER**, which is calculated from a detection result of the position detection sensors **S11** and **S12**. At this time, the controller CU sets a movement amount and a movement direction in the correction and movement of the transfer robot **TR1** based on the acquired misalignment amount and the acquired misalignment direction, and corrects the movement of the transfer robot **TR1**. Accordingly, even if a misalignment occurs in the edge ring **ER**, the substrate processing system **PS** may smoothly load the edge ring **ER** into the ring accommodation module **RSM**.

[0197] In addition, the transfer method in FIG. **19** may be applied to even a case where the cover ring **CR** is unloaded from any one of the processing modules **PM1** to **PM7** to the vacuum transfer module **TM** and transferred into the ring accommodation module **RSM**. Further, the transfer method in FIG. **19** may be applied to even a case where the upper ring **222** or the lower ring **221** in FIG. **11** is unloaded from any one of the processing modules **PM1** to **PM7** to the vacuum transfer module **TM** and transferred to the load lock module **LLM**. Further, the transfer method in FIG. **19** may be applied to even a case where the upper ring **222** or the lower ring **221** in FIG. **11** is unloaded from any one of the processing modules **PM1** to **PM7** to the vacuum transfer module **TM** and transferred to the ring accommodation module **RSM**.

[0198] Alternatively, as another modification, the substrate processing system **PS** and the transfer method may detect a position of the edge ring by the position detection sensor **S1** or **S2**, by raising the edge ring **ER** and then lowering the edge ring **ER** by the supporting pins **521**, to place the edge ring **ER** on the ring support surface **112b**. If a position misalignment amount of the edge ring **ER** is smaller than a threshold value, the substrate processing system **PS** and the transfer method may perform an unloading processing from the processing module **PM1** (unloading while detecting a position of the edge ring **ER** by the position detection sensors **S11** and **S12**, or the like) by raising the edge ring **ER** again by the supporting pins **521** and transferring the edge ring **ER** to the transfer robot **TR1**, and perform a loading processing into the ring accommodation module **RSM** (correction and movement based on the position misalignment amount). In the unloading processing of the edge ring **ER** from the processing module **PM1** and the loading processing of the

edge ring ER into the ring accommodation module RSM, the processing flow shown in FIG. 19 may be performed. Meanwhile, when the position misalignment amount of the edge ring ER is the threshold value or more, the substrate processing system PS and the transfer method may perform a processing of taking out the edge ring ER by issuing an alarm and opening the plasma processing chamber 10 in an atmosphere.

[0199] The embodiment disclosed in the above includes, for example, the following aspect.

(Supplementary Note 1)

[0200] A substrate processing system including: [0201] a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring; [0202] a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and [0203] a controller, [0204] wherein the controller performs: [0205] (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; [0206] (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and [0207] (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

(Supplementary Note 2)

[0208] The substrate processing system of Supplementary Note 1, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor included in the transfer robot.

(Supplementary Note 3)

[0209] The substrate processing system of Supplementary Note 2, wherein, in step (B), the controller acquires a gap amount of the ring in a horizontal direction with respect to a substrate support surface of the substrate support based on a position of the substrate support surface and the position of the ring, which are detected by the position detection sensor.

(Supplementary Note 4)

[0210] The substrate processing system of any one of Supplementary Notes 1 to 3, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, a leakage amount of a gas by supplying the gas between the support surface of the substrate support and a rear surface of the ring in a state in which the ring is electrostatically attracted to the support surface.

(Supplementary Note 5)

[0211] The substrate processing system of any one of Supplementary Notes 1 to 4, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, image information of the ring, which is imaged by a camera included in the transfer robot.

(Supplementary Note 6)

[0212] The substrate processing system of any one of Supplementary Notes 1 to 5, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor installed in the vacuum transfer module, when unloading the ring from the processing module by the transfer robot.

(Supplementary Note 7)

[0213] The substrate processing system of any one of Supplementary Notes 1 to 6, wherein, if it is determined to perform the correction in step (C), the controller further performs correction and movement of moving the ring according to the position misalignment amount by the transfer robot, and [0214] wherein, if it is determined not to perform the correction in step (C), the controller further performs: transferring the ring by the transfer robot, without performing the correction and movement.

(Supplementary Note 8)

[0215] The substrate processing system of Supplementary Note 7, wherein, if it is determined to perform the correction in step (C), the controller performs the correction and movement by the

transfer robot in the processing module.

(Supplementary Note 9)

[0216] The substrate processing system of Supplementary Note 8, wherein, in the correction and movement, the controller sequentially performs: [0217] receiving the ring in which a misalignment occurs from the lifter by the transfer robot; [0218] moving the transfer robot according to the position misalignment amount; and [0219] transferring the ring from the transfer robot to the lifter.

(Supplementary Note 10)

[0220] The substrate processing system of Supplementary Note 8, wherein, in the correction and movement, the controller performs: moving the transfer robot according to the position misalignment amount before receiving the ring from the lifter by the transfer robot.

(Supplementary Note 11)

[0221] The substrate processing system of Supplementary Note 7, wherein, if it is determined to perform the correction in step (C), the controller further performs: transferring the ring to a ring accommodation module configured to accommodate the ring while performing the correction and movement on the ring in which a misalignment occurs by the transfer robot.

(Supplementary Note 12)

[0222] The substrate processing system of any one of Supplementary Notes 1 to 3 and 5, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is placed on the substrate support by lowering the lifter.

(Supplementary Note 13)

[0223] The substrate processing system of any one of Supplementary Notes 1 to 3 and 5, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is raised by the lifter in step (A).

(Supplementary Note 14)

[0224] The substrate processing system of any one of Supplementary Notes 1 to 13, further including: a first ring accommodation module connected to the vacuum transfer module and configured to accommodate the ring, [0225] wherein the controller further performs: loading the ring unloaded from the processing module into the first ring accommodation module by the transfer robot.

(Supplementary Note 15)

[0226] The substrate processing system of any one of Supplementary Notes 1 to 13, further including: [0227] an atmospheric transfer module connected to the vacuum transfer module via a load lock module; and [0228] a second ring accommodation module connected to the atmospheric transfer module and configured to accommodate the ring, [0229] wherein the controller further performs: loading the ring unloaded from the processing module into the second ring accommodation module via the load lock module and the atmospheric transfer module by the transfer robot.

(Supplementary Note 16)

[0230] The substrate processing system of any one of Supplementary Notes 1 to 15, wherein the processing module performs a substrate processing in a state in which the ring is electrostatically attracted, and [0231] wherein, before step (A), the controller further performs: removing charges from the ring.

(Supplementary Note 17)

[0232] The substrate processing system of any one of Supplementary Notes 1 to 16, wherein the controller further performs: cleaning of an interior of the processing chamber by generating plasma during step (A) or before and after step (A).

(Supplementary Note 18)

[0233] The substrate processing system of Supplementary Note 17, wherein the cleaning of the interior of the processing chamber during step (A) or after step (A) is performed in a state in which the ring is spaced apart from the support surface of the substrate support.

(Supplementary Note 19)

[0234] The substrate processing system of any one of Supplementary Notes 1 to 18, wherein the lifter includes a supporting pin and an actuator configured to vertically move the supporting pin.

(Supplementary Note 20)

[0235] A transfer method of unloading a ring disposed at a periphery of a substrate from a processing module by a transfer robot of a vacuum transfer module connected to the processing module, wherein the processing module includes a processing chamber, a substrate support configured to support the substrate and the ring in the processing chamber, and a lifter configured to raise and lower the ring, the transfer method including: [0236] (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; [0237] (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and [0238] (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

[0239] In addition, the present disclosure is not limited to the configurations shown herein, including configurations exemplified in the embodiments, combinations with other elements, and the like. In this regard, this may be changed without departing from the spirit of the present disclosure, and may be properly set according to an application form thereof. Further, in items described in a plurality of embodiments, other configurations may be used within a non-contradictory range and may be combined within a non-contradictory range.

[0240] For example, in the embodiment, a capacitively coupled plasma apparatus is described as an example. However, the present disclosure is not limited thereto, and may be applied to other plasma apparatuses. For example, instead of the capacitively coupled plasma apparatus, an inductively coupled plasma (ICP) apparatus may be used. In this case, the inductively coupled plasma apparatus includes an antenna and a lower electrode. The lower electrode is disposed in a substrate support, and the antenna is disposed on or above a chamber. Further, an RF generator is coupled to the antenna, and a DC generator is coupled to the lower electrode. Therefore, the RF generator is coupled to an upper electrode of the capacitively coupled plasma apparatus or the antenna of the inductively coupled plasma apparatus. That is, the RF generator is coupled to the plasma processing chamber **10**.

[0241] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-174836, filed on Oct. 31, 2022, the entire contents of which are incorporated herein by reference.

[0242] According to the present disclosure in some embodiments, it is possible to suppress interference between a ring and other parts when transferring the ring.

[0243] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A substrate processing system comprising: a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring; a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and a controller, wherein the controller performs: (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; (B) after step (A), acquiring

an index related to a position misalignment amount of the ring; and (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

2. The substrate processing system of claim 1, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor included in the transfer robot.

3. The substrate processing system of claim 2, wherein, in step (B), the controller acquires a gap amount of the ring in a horizontal direction with respect to a substrate support surface of the substrate support based on a position of the substrate support surface and the position of the ring, which are detected by the position detection sensor.

4. The substrate processing system of claim 1, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, a leakage amount of a gas by supplying the gas between the support surface of the substrate support and a rear surface of the ring in a state in which the ring is electrostatically attracted to the support surface.

5. The substrate processing system of claim 1, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, image information of the ring, which is imaged by a camera included in the transfer robot.

6. The substrate processing system of claim 1, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor installed in the vacuum transfer module, when unloading the ring from the processing module by the transfer robot.

7. The substrate processing system of claim 1, wherein, if it is determined to perform the correction in step (C), the controller further performs correction and movement of moving the ring according to the position misalignment amount by the transfer robot, and wherein, if it is determined not to perform the correction in step (C), the controller further performs: transferring the ring by the transfer robot, without performing the correction and movement.

8. The substrate processing system of claim 7, wherein, if it is determined to perform the correction in step (C), the controller performs the correction and movement by the transfer robot in the processing module.

9. The substrate processing system of claim 8, wherein, in the correction and movement, the controller sequentially performs: receiving the ring in which a misalignment occurs from the lifter by the transfer robot; moving the transfer robot according to the position misalignment amount; and transferring the ring from the transfer robot to the lifter.

10. The substrate processing system of claim 8, wherein, in the correction and movement, the controller performs: moving the transfer robot according to the position misalignment amount before receiving the ring from the lifter by the transfer robot.

11. The substrate processing system of claim 7, wherein, if it is determined to perform the correction in step (C), the controller further performs: transferring the ring to a ring accommodation module configured to accommodate the ring while performing the correction and movement on the ring in which a misalignment occurs by the transfer robot.

12. The substrate processing system of claim 1, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is placed on the substrate support by lowering the lifter.

13. The substrate processing system of claim 1, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is raised by the lifter in step (A).

14. The substrate processing system of claim 1, further comprising: a first ring accommodation module connected to the vacuum transfer module and configured to accommodate the ring, wherein the controller further performs: loading the ring unloaded from the processing module into the first ring accommodation module by the transfer robot.

- 15.** The substrate processing system of claim 1, further comprising: an atmospheric transfer module connected to the vacuum transfer module via a load lock module; and a second ring accommodation module connected to the atmospheric transfer module and configured to accommodate the ring, wherein the controller further performs: loading the ring unloaded from the processing module into the second ring accommodation module via the load lock module and the atmospheric transfer module by the transfer robot.
- 16.** The substrate processing system of claim 1, wherein the processing module performs a substrate processing in a state in which the ring is electrostatically attracted, and wherein, before step (A), the controller further performs: removing charges from the ring.
- 17.** The substrate processing system of claim 1, wherein the controller further performs: cleaning of an interior of the processing chamber by generating plasma during step (A) or before and after step (A).
- 18.** The substrate processing system of claim 17, wherein the cleaning of the interior of the processing chamber during step (A) or after step (A) is performed in a state in which the ring is spaced apart from the support surface of the substrate support.
- 19.** The substrate processing system of claim 1, wherein the lifter includes a supporting pin and an actuator configured to vertically move the supporting pin.
- 20.** A transfer method of unloading a ring disposed at a periphery of a substrate from a processing module by a transfer robot of a vacuum transfer module connected to the processing module, wherein the processing module includes a processing chamber, a substrate support configured to support the substrate and the ring in the processing chamber, and a lifter configured to raise and lower the ring, the transfer method comprising: (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).
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